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# **STYLE MODULE OVERVIEWS**

Version 3.5.2

## **Product Guide**



# FRONTISPIECE

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## Glossary



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## Chapter 1

### Style Sales Analysis Overview



# Introduction

In today's increasingly competitive markets, the availability of clearly presented and easily manipulated sales information is of paramount importance in directing the efforts of the business.

The Style Sales Analysis application enables you to produce your own views of sales information, to enable you to make comparisons and identify trends which are vital to strategic planning.

One of the great strengths of Sales Analysis is the extent to which each individual user is able to tailor the system to their own requirements without the need for any programming. This flexibility starts with the Sales History Database which is defined by the user, along with the level at which Sales History is held. The formats of enquiry are chosen by the user, and a new view of the information may be created quickly and easily without the need for data processing personnel.

Reports may be created via a simple series of prompts, and may be requested on a regular basis, or may be simple ad-hoc enquiries to answer a pressing question.

## Relationship To Other Style Applications

The Sales Analysis application makes use of primary data held in the Inventory Management and Sales Order Processing applications within Style.

It is within Inventory Management that details of items are held, including data such as item type, item group, division and sub-division.

Inventory Management also contains a database of codes and descriptions, used to maintain an abbreviated code for such details as item group, or salesman, together with a meaningful description. The great benefit of the description database is that it enables a consistent and easily validated use of codes, and also provides for a meaningful description at report or enquiry time.

Customer details are held within the Sales Order Processing application, including data such as salesman, territory, and customer class.

## Company Profile

As with all Style products, Sales Analysis may be used by a number of companies, each with differing requirements. A profile of each company using the system is maintained, which details such things as the name of the company and the level at which sales history is to be held.

## Calendar

To maintain independence from financial or manufacturing time periods, Sales Analysis has its own calendar, which holds the dates on which sales weeks or periods end. This enables Sales Analysis to allocate each sale to the appropriate period, according to its date.

## Budgets

Budgets may be defined within Sales Analysis, and have two main characteristics, a type and a format.

The type is an overall classification, such as 'Budget', 'Target' or 'Forecast'. The format describes the attribute or attributes to which the budget relates, for example, it is possible to have a 'Target' for Salesman, salesman being a possible item on the Sales History Database, or a 'Forecast' for Customer account by Item group.

Budgets are entered by Sales History Period for a type and format previously defined. Spread profiles may be used to aid the creation of budgets, allowing an annual total to be spread over the years sales periods according to a known pattern.

Sales Analysis reports enable the comparison of budgeted values with actual sales.

# The Sales History Database

Fundamental to Sales Analysis is the Sales History Database, designed to enable a high level of tailoring to individual requirements, without programming changes.

When the system is installed, you are able to choose the customer and item data elements which you wish to hold on the database. In addition to the item code, customer code and sales period, you may choose six data elements relating to the Inventory Management item master, and six relating to the Sales Order Processing customer master file. Quantity, Cost and Sales Value are also held, with provision for four further values.

You may also choose to change the screen and report headings for the data elements held on the database. For example, if you use Item type, to indicate whether an item is 'made in' or 'bought out', you can change the heading to say just that. This change is then fully incorporated within the database itself, allowing full utilisation of utilities, which will print your chosen column headings.

The Sales Analysis database is held at style level rather than full product (style/colour/size/fit).

# Updates And Processing

## Sales History Update

Actual sales history is drawn from the Sales Order Processing invoice data. The Sales History Update function takes this data and adds it to the Sales History Database, using the calendar to determine the appropriate sales period.

This may be run as often as required, the actual frequency depending upon how up-to-date the sales history needs to be.

Sales history is maintained at style level. A full product sales history database is available for creation of your own query reports.

## Sales History Refresh

Sales Analysis enables you to selectively refresh customer or item data on the database in the event of a change.

For example, if sales regions are re-organised, so that customers fall into different regions, after the customer data in Sales Order Processing has been updated, it is possible to use the Sales History Refresh function to bring the Sales History Database into line with this re-organisation, and allow analysis to show past sales to customers, within their new regions.

## Periodic Updates

Whilst Sales Analysis has no pre-defined 'month-end' reports, the functions are provided to allow you to run whatever regular reporting is required.

As you create your own reports, you are prompted to specify that they are to be run on a regular basis, for example daily, or at month end. The daily, weekly, period-end, and year-end update functions select those reports you have requested, and produce them at the appropriate time.

These regular updates also maintain the current period and year, held on the Company Profile.

# Enquiries

Sales Analysis provides a powerful enquiry function, which enables you to create a view of any data element on the database, analysed in terms of any other data item. The values displayed may also be chosen from the values on the database, and the time periods may be defined in a variety of different ways.

A new view may be created simply by asking for an analysis which has not previously been used, and the application will then offer to create it.

The Sales Enquiry provides you with four columns of values, chosen by means of a number of formats, created at installation time.

A format specifies the time periods to be displayed, together with the particular value to be displayed.

For example:

This period

Last period

3 periods ago

4 periods ago

Value to be shown=Sales Value.

OR:

This period

Last period

This year

Last year

Value to be shown=Quantity.

To enquire on sales history, in a chosen format, you must enter two items:

- The data element on which to enquire
- The data element by which to analyse

For example, enquire on salesman 01 by item group, would show a breakdown by item group of salesman 01's sales. The enquiry may then be advanced to show the next salesman's sales by item group, and so forth.

Options are provided to invert the display, in order to show the sales of a selected item group by all salesmen.

A bar chart function is also provided to provide a graphical representation of the data.

# Reports

Sales Analysis contains report generator functions, which enable you to specify your own reports simply and quickly. You can produce summaries at several levels, for example, a report showing item sales summarised by salesman; or produce a ranking of the best selling item groups comparing one sales period with another.

A report may be created to run on a regular basis, perhaps at month-end, or a report may be used once to answer an ad-hoc enquiry.

In addition to this there are a number of standard Housekeeping type reports.

## Item File Print

This provides a listing of the items defined within Inventory Management, together with their attributes chosen for inclusion on the Sales History Database.

## Customer File Print

This provides a listing of the Customers defined within Sales Order Processing, together with their attributes chosen for inclusion on the Sales History Database.

## Report Specification Print

This function provides a listing of the specifications of reports defined within Sales Analysis.

## Sales Report

The Sales Analysis system provides functions which enable you to run an existing sales report, or to create and optionally run a new or ad-hoc report.

Such is the ease of creating a report that Sales Analysis is particularly adept at providing reports for ad-hoc queries.

A report is created using a series of prompts by means of which you specify the reports sequence, selection and content, and when the report is to be run.

The report specification may be saved, and is then available to be run regularly, or on request.

## Ranking Analysis

The Ranking Analysis function within Sales Analysis is used to sequence or rank elements by the results of a summarised value, and optionally, make comparisons between one period and another.

Thus, you could for example, choose to rank salesmen by sales value, which would summarise their sales values in order. You might choose to make a comparison and compare this month's ranking with last months, to see what changes have taken place.

The ranking function may also be made to work at a second level, and rank one element within another, for example, item group, within region, showing the best selling item groups for each region.

# Utilities

Sales Analysis contains a set of utility functions which are intended for use at system installation time or under certain special circumstances.

These utilities enable ease of software installation, and continued maintenance.

## Pre Requisites

This section deals with the steps involved in implementing Style Sales Analysis, from the initial installation to the production of the first enquiries and reports.

Sales Analysis may be installed as an additional module within an existing Style environment, or may be part of a new installation.

The logical pre-requisites are the same in both cases.

As with other products, Sales Analysis is supported by System Manager, which must be installed first.

Sales Analysis also requires that Inventory Management and Sales Order Processing are installed. From Inventory Management it obtains details about the items which are sold and from Sales Order Processing it obtains details of customers, and invoice details representing sales. Inventory and Sales Order Processing are part of your standard Style application.

In order for Sales Analysis to extract information about sales, the item details within Inventory Management must be maintained such that the information for analysis is present, for example item group. Likewise, the customer details required for analysis must also be present within Sales Order Processing, and some invoices must have been printed.

## Basic Data

Before any processing can take place within Sales Analysis, some basic details must be set up within the application.

## Company Profile

A Company Profile must be created for each company which is to use the application. This contains processing details specific to each company, and also the current year and period, maintained by period-end routines.

## Calendars

Sales Analysis provides information grouped by sales period, whereas invoices are created as individual dates. In order for invoices to be placed in the appropriate period, a calendar is used.

Calendars must be created before processing starts. If you choose to hold history at a weekly level, calendars should be created for both week and period for all years which you intend to process.

## Sales History Time Period Selection

The information shown on the Sales Analysis enquiry is determined by the creation of formats.

These define the time periods and the values which will be shown, and must be created before any attempt is made to use the enquiry function of Sales Analysis.

A default format 1 is used by the application, and therefore, this at least must be created.

# Database

Central to the Sales Analysis application is its database, which must be defined before processing is to start.

## Change Initial Field Descriptions

Within Sales Analysis, data elements or fields have internal names, which cannot be altered, but also external names by which they are identified to users of the system. The Change Initial Field Descriptions function may be used to step through the data elements, and change the names or descriptions of these in order to suit a particular installation.

## Define Database

The database must be created before processing can begin. You should consider the data elements which you wish to hold before you create the database.

# Budgets

Within Sales Analysis you may choose to maintain budgets, and in reports, compare budgets with actual sales. You may choose to run the application without budgets initially and introduce them later. A budget is defined in terms of a budget format, and a budget type.

You may also create spread profiles, which aid budget entry, if sales fall into a known pattern.

## Definition

In order to describe the data element(s) to which a budget applies, a format must be created. This also determines the kind of value to be held, for example, cost value or sales value.

The Budget format function should be used to create the formats you require.

## Spread Profiles

The Spread Profiles function enables you to define the pattern of sales over a year, by period. A number of profiles may be created and used to aid budget entry.

## Budget Entry

Budgets may be entered for a year, against the data elements defined by the budget format, for the values also specified by the format.

Spread profiles may be used to spread an annual total over all the periods in a year.

# Sales History

Sales history enables invoice analysis and the creation of enquiry, reporting and updating functions based upon those data elements that are relevant to your needs. For more information see:

- Invoices
- Updating Sales Analysis
- Enquiry
- Reporting

## Invoices

Sales Analysis draws its information on sales from data within Sales Order Processing. For processing to start, invoices must have been printed within Sales Order Processing.

## Updating Sales Analysis

The Sales History Update function takes printed invoice data and posts the information to the Sales History Database. Once posted, the invoices are marked as such and will not be included in subsequent updates. If large volumes of data are involved this function may take some time to complete.

## Enquiry

The Sales Enquiry function will prompt you to enter a data element on which to enquire, and to select a data element by which to analyse. The first time a new enquiry combination is selected, the system will build summary data but there after the enquiry will be performed straight away.

## Reporting

Sales Analysis enables you to create your own Sales Reports and Ranking Analysis Reports.

The Reports Specifications and Ranking Analysis Specification functions allow you to define reports quickly and easily. They may be created and run as part of the specification maintenance, or they may be selected by name at a later time, or included in the regular update runs.

## Miscellaneous

For more information regarding Sales Analysis file library details and Enquiry bar chart characters see:

- [Library For User Created Objects](#)
- [Bar Chart Characters And Colours](#)

## Library For User Created Objects

Sales Analysis creates the system program and file objects during its execution. These must be directed to a specific library. A data area called SAUSRLIB contains the name of the library in which these objects will be placed. The default supplied by the system will be the Sales Analysis file library.

## Bar Chart Characters And Colours

The characters to be used in the Enquiry bar chart are contained in the data area SABARCHARS, in hexadecimal form, and the display attributes are held in the data area SABARCHCOL. These both contain default values, which you may change to tailor the bar chart display.

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## Chapter 2

# Style Purchase Management Overview



# Introduction

A comprehensive Purchase Management system is necessary for controlling a cost effective and accurate supply of items. The Style Purchase Management application provides the necessary information and facilities in the following key areas:

- Order creation
- Order follow-up
- Container tracking
- Goods receiving
- Quality assurance
- Invoice matching
- Performance analysis

The application may be controlled and tailored to each company's requirements by use of profiles, parameters and tables. Any number of companies may be defined.

As organisations expand into international markets, multi-currency handling is essential and is integrated into this application.

## Relationship To Other Applications

The Purchase Management application is a fully standardised Style application and operates under the control of System Manager.

The application is designed to be used in conjunction with the following Style applications:

- Inventory Management
- Purchase Ledger
- Purchase Requisitioning
- Warehousing

The Company Profile is used to identify the applications interfaced.

## Inventory Management

Purchase Management is fully integrated with the Inventory Management application in an on-line mode.

'Companies' communicate with each other on a one-to-one basis, identified by the Company Code.

Where Inventory Management is installed, Purchase Management uses the Inventory Management Item Master for the specification of items. On order and physical stock balances are maintained by Purchase Order Entry and Receiving functions.

Purchase Management writes inventory movements, and maintains actual cost details in the Inventory Management application.

## Purchase And General Ledgers

The Style Purchase Management application is fully integrated, on-line, with the Purchase Ledger. The integration of Purchase Management, Purchase Ledger and General Ledger is controlled as follows:

- It is possible for many Purchase Ledger companies to interface to one General Ledger company. The Purchase Ledger Company Profile defines the General Ledger company to which the Purchase Ledger company interfaces.
- The Purchase Management Company Profile defines a General Ledger Company Code for 'interface' purposes. The Purchase Management company will interface with any Purchase Ledger company which links to this General Ledger. As a result, a single Purchase Management company may link to many Purchase Ledger companies.

Purchase Management uses the standard Purchase Ledger supplier data and provides facilities for matching receipts with Purchase Ledger items.

## Purchase Requisitioning

The Purchase Management application is fully integrated with the Purchase Requisitioning application.

Companies communicate with each other on a one-to-one basis, identified by the company code.

The Purchase Requisitioning application shares Master File data with and creates order details in the Purchase Management application.

## Warehousing

The Purchase Management application performs internal checks to determine:

- If the Warehousing application is installed.
- If the Company/Stockroom being processed is defined to the Warehousing application.

If so, when goods are received or transferred into stores a window is displayed for Warehousing specific information to be entered. This includes packing code, and expiry date (for consumables). These items are then automatically processed by the put-away routines in Warehousing, which decide where the items are to be located.

# Configuration

As with all Style applications, Purchase Management can be operated for a number of companies, the characteristics of each being maintained on the Company Profile.

The primary information maintained on the Company Profile is:

- Application set-up parameters
- Currency defaults
- Output queues for reports
- Interfaces to other applications
- Order numbering
- Goods Received Note (GRN) numbering
- Debit note numbering
- Purchasing officer number range
- Item class range
- Requisitions system options

These default values improve efficiency of data entry, provide the parameters for the generation of document references, provide specification of the technical environment, and enable set up of ranges for data integrity checking.

# Maintenance Of Reference Data

Purchase Management uses Master File data set up in other Style applications. However, there are some Master Files which are set up specifically for the application.

Information for the following may be maintained:

- Item Specifications
- Purchasing Officer names
- Item Supplier Profiles
- Delivery Instructions
- Receiving Addresses
- Supplier External text
- Parameters and Sundry Codes
- User Authorities
- Container details

## Item Specifications

Where Style Inventory Management is not installed, item specifications are set up in Purchase Management. These specifications detail classification, description, associated nominal account codes, unit of measure details, VAT Code and alternative/supersession data. Text may also be maintained for the item, which may be optionally added to the purchase order line in order entry.

## Purchasing Officer Names

Detail is held of the company's officers and/or planners who use this application. This is then recorded against each order, and is a displayed item on the application's enquiries.

## Item Supplier Profiles

For each supplier of an item, an item/supplier profile is required. The profile includes price, ordering rules, lead time and supplier reference details.

A cumulative total of orders placed and receipts are also maintained by the application for each item/supplier profile.

## Delivery Instructions

A set of delivery instructions may be maintained for each company. Each set of instructions is identified by a three character code and comprises up to three lines of text.

The delivery instruction code can be selected at order entry, and is then subsequently printed on the purchase order form.

## Receiving Addresses

A set of receiving addresses may be maintained for each company. Each one comprises a name, and up to five lines of address.

The code may be selected at order entry, to specify the address to which the goods are to be sent. Otherwise, it defaults to the receiving address code set on the Company Profile.

## Supplier External Text

For each supplier, up to 99 lines of text may be maintained. At order entry, this text can optionally be selected for printing on the order.

## Parameters And Sundry Codes

A number of parameters should be maintained for each company. These are used to control the application and to tailor it to the company's requirements. These parameters include:

- Unit of measure codes
- Item Classifications
- Order Classifications
- Scrap Reason Codes
- Schedule Order Types
- User Defined Fields
- Warehouse Codes

## User Authorities

Access to Purchase Management procedures is controlled by the standard authorisation features of System Manager. The application does provide an additional level of user authorisation within certain procedures.

Authority levels may be set for standard user profiles. These authority levels limit the ability of the user to:

- Under receive an order and specify that the order is complete
- Over receive an order
- Match a purchase invoice at a price greater than that specified for the order

The authority levels are set as absolute values, or as percentages of the relevant order line value.

## Container Details

Any number of containers may be maintained for each company. Each one comprises of data such as shipping information, tariff codes, weight, letter of credit, receiving port and expected dates. A purchase order is then attached to a container before goods receipt can be actioned.

# Processing

The normal day-to-day processing in Purchase Management can be divided into three types of tasks:

- Creating Orders.
- Receiving Goods.
- Matching invoices to goods received.

## Creating Orders

Purchase orders can be created by two different methods:

By generating purchase orders from requisitions via the facility in the Requisitions application.

OR

By entering the purchase order directly into the Purchase Management application via Order Entry.

## Order Creation From Purchase Requisitioning

Purchase requisitions may be created from any of the applications shown below, and then transferred into purchase orders.

- Inventory Management
- Distribution Requirements Planning
- Purchase Requisitioning
- Purchase Management
- Material Requirements Planning
- Manual Entry

The creation of purchase orders from Purchase Requisitioning may be done selectively for any of the following:

- Release flag
- Supplier
- Purchasing Officer
- Order Class
- Stockroom

Separate purchase orders can optionally be created on change of supplier and/or purchasing officer. The rules governing this are specified on the Purchase Management Company Profile.

## Order Entry

The entry of a purchase order can be broken down into three stages - Header, Items and Options.

When entering the Order Header, the following information can be specified as:

Order Class

Specifies if the order is for stocked items, non-stocked items, or services.

Order Type

Defines whether the order is of a normal (single delivery per line), scheduled, or blanket type.



*Note: A 'Scheduled' order is a firm commitment with a supplier for quantities to be delivered on specified dates. Multiple deliveries are expected for each line. A 'Blanket' order is similar to a scheduled order but the commitment to quantities and dates is only provisional. It is processed in the same way as a scheduled order within this application.*

Other details specified on the order header are:

- Order Date
- Order Number
- Purchasing Officer
- Department
- Discount
- Job Number

When adding items to the order, the standard Inventory Management alpha search is available for identifying and selecting the correct item.

The supplier's item reference, unit of measures, price and discounts are defaulted for the item, but may be changed. However, for non-stocked items there are no defaults for price or units.

Upon entry of the style number, a matrix is displayed to allow quantity entry against individual colour \ size options. Nominal cost code, job number and department may also be entered for each item or style.

Options are available when completing each order to:

- Enter further orders or return to the menu.
- To print or fax the order.
- To use existing external supplier text or not.
- To add new text for printing on the order, i.e. external text.
- To add new text for display only, i.e. internal text.

- To cancel the whole order.

Full amendment facilities are available on outstanding orders, and an audit trail is provided of amendments made.



*Note: New orders are detailed on the New Orders Audit report. This list can act as an authority document prior to printing the order. Further, this audit must be produced prior to receipts being processed against the order.*

## Receiving Goods

All goods or services received must be covered by a purchase order.

To enable the correct purchase order to be processed, all outstanding orders by item or supplier can be viewed. Furthermore, if the item code and supplier code are unknown, searches are available for these. Obviously, the purchase order number or container number may be entered directly, if known.

Goods may be received in up to three locations:

- Goods Inwards
- Inspection
- Stores

For stock items, Inventory Management is only updated when receipt (or transfer) is made to stores. Items may be subsequently moved from their original receipt location using the Transfer Goods Location option in this application.

In recording the receipt, the whole order may be received in its entirety. Alternatively, the receipt can be done line by line, with partial receipts allowable.

The user's authority is checked before under/over delivery can be accepted.

If the receipt is made into stores location and the Warehousing application is installed, a window is displayed for entry of package code (and expiry date for consumables).

Both internal order and order line text may be updated at the receiving stage.

The Goods Receipt Note (GRN) may be printed along with labels if required. GRN numbers may be automatically generated. These GRN options are specified on the Company Profile.

Once the goods have been received, the following facilities within this application can be used to attain high levels of Quality Assurance:

- Transfer of goods into or out of inspection
- Return of unwanted goods, with optional replacement
- Scrapping of faulty goods, with analysis by Reason Codes
- Optional creation of rejection notes
- Ability to match a credit note to a return, at a later date

## Invoice Matching

Invoice matching is designed to close the loop between the Purchase Ledger application and the Purchase Management application by providing matching of receipts with invoices or credit notes on the Purchase Ledger Log.

The Invoice Matching routine performs two basic functions:

- Receipts within the Purchase Management application are flagged as having been matched with invoices. The pricing details are stored in Purchase Management for future analysis in the supplier performance and price variance reports.
- Invoice Log records in the Purchase Ledger are flagged as matched, and dissections to the appropriate nominal codes are set up.

Invoices, credit notes and journals on the Invoice Log may be matched against any number of advice notes and parts of advice notes.

A report of outstanding advice notes by supplier is available to assist Invoice Matching.

Once invoices have been matched, there is a separate facility in the Purchase Ledger which must be used to transfer the invoice from the Log to the Purchase Ledger proper. A report of matched invoices (on the Invoice Log) is available to trigger release of these invoices into the Purchase Ledger.

Text may be entered, reviewed and amended against the Invoice Reference at the time of matching.

An audit trail of Invoices Matched may be printed if required.

Where the Invoice Log is not in use, receipts within Purchase Management may be matched using the 'stand-alone' matching procedure. This procedure is similar to standard matching, but involves no interface to the Purchase Ledger Log.

## Interface To General Ledger

Purchasing interfaces to JBA General Ledger can be made in the following ways:

- Goods Receipt/Adjustment
- Match Invoice to Receipt

## Goods Receipt

At this point stock is updated within the General Ledger with a contra-posting to the accrual accounts. The account codes used are defined against the item within inventory.

In addition any landed costs are posted separately to the General Ledger at receipt stage.

## Invoice Match

This function will back out the postings for stock and accruals made at goods receipt stage. These postings are replaced by postings for stock, debtors and price variance.



*Note the back out of stock and accruals is done directly from invoice match but the replacement postings for stock, debtors and price variance do not update the general ledger until the purchase ledger log is posted to the General Ledger.*

# Enquiries

Enquiries may be grouped into four categories:

- Outstanding Order Searches
- Order Details
- Sources of Supply
- Financial

## Outstanding Order Searches

Multiple routes are provided for identifying an outstanding purchase. Enquiries exist by:

- Item
- Supplier
- Job Number
- Supplier Item Reference
- Purchasing Officer
- Container

## Order Details

Once the relevant purchase order has been identified, an enquiry exists to display details of that order. Each line can then be selected to show receipts to date along with other line details.

## Sources Of Supply

Enquiries are available to satisfy these questions:

Which suppliers supply this item?

Which items have been bought from this supplier?

These enquiries are known as 'Where bought?' and 'What bought?' respectively.

## Financial

An enquiry exists to display the details of an invoice, including the value matched and matched receipt details. Also available is an enquiry on outstanding orders for a given Cost Code.

# Reports

As part of the general policy of Style applications, before a report is run selection and sequencing parameters are normally available.

In general, standard reports are printed on standard listing paper with 132 print positions and six lines to the inch. In Purchase Management the printing of orders, goods receipt notes and labels are made on special stationery.

The reports in Purchase Management can be grouped into one of six categories:

- Actions
- Status
- Analysis
- Audits
- File Lists
- Exceptions

## Actions

These reports are used to trigger actions to be taken, and are necessary to progress an order through the application:

- New Orders Report
- Print Orders
- Print GRNs/labels

## Status Reports

These show the current position of purchase orders within the application:

- Order Pre-print
- Deliveries Due
- Order Confirmation
- Advice Notes not Invoiced
- Accruals (Receipts not Invoiced)

## Analysis

A selective overview of the Purchase Management function, and also how effectively it performs, is provided by:

- Receipts and returns
- Goods Value by Location
- Price Variance
- Supplier Performance
- What bought
- Where bought
- Order Value by Cost Code
- Purchase Commitment
- Container

## Audits

The following audits are provided for processing functions in Purchase Management:

- Order Amendments
- Receipts and Returns

## File Lists

Full file listings may be requested for the following:

- Item Specifications
- Purchasing Officers
- Receiving Addresses
- Delivery Instructions
- Parameters and Sundry Codes
- User Authority Values

## Exceptions

These include discrepancies within the database which may need actions to rectify.

- Item/Supplier Profile exception report
- Order File exception report

# Pre-Requisites

## Applications

For the basic operation of Purchase Management, only System Manager is pre-requisite.

However, the application is designed to be used in conjunction with the following:

- Inventory Management
- Purchase Ledger

If the Style Inventory Management application is not installed, then all purchased items must be maintained within Purchase Management.

If the Purchase Ledger is not fully installed, then at least the Supplier Master and Currency Rate data of the Purchase Ledger must be maintained.



*Note: If the Purchase Orders are to be sent directly to the Supplier via a Facsimile transmission, then the Fax module is a pre-requisite (This is called Cobwebb and is only applicable to the AS/400 systems). This handles the actual transmission of the fax once it has been created via Purchase Management, and is available from your software support organisation.*

## Data

Prerequisite data includes supplier definition, currency rate set up and item definition. For more information see Supplier, Currency and Items.

## Supplier

The Suppliers to be referenced in Purchase Management must be defined on the Supplier Master File. This is maintained within the Purchase Ledger.

## Currency

If applicable, current rates for all relevant currencies must be set up, using the General Ledger functions.

## Items

All items to be included in Purchase Management must be defined to Style Inventory Management, if installed.

Otherwise, the item must have a specification defined within Purchase Management.

# Application Configuration

This describes the way you wish to set up various profiles within the application. For more information see:

- Companies
- Item Specifications
- Purchasing Officers
- Item Supplier Profiles
- Delivery Instructions
- Receiving Addresses
- Supplier External Text
- User Authorities
- Parameters And Sundry Codes
- Fax Numbers

## Companies

For each company to be processed within Purchase Management, a profile record must be maintained within the application.

The primary information maintained per company is:

- Company name
- Default Supplier Address Code
- Default Currency Rate Codes
- Output Queues for Reports
- Interfaces to other Applications
- Order Numbering
- Goods Receipt Note (GRN) Numbering
- Debit Note Numbering
- Purchasing Officer Number Range
- Requisitions Options, (where Requisitions application is installed)
- Default Goods Receiving Booking-In-Point
- Receipt Time Fence Number of Days
- Dummy Items, (for Delivery Instructions)
- Dummy Supplier, (for Receiving Addresses)

Within each company, the Reference Data described in the remainder of this section must/may be maintained.

## Item Specifications

Each item must have a specification within the Purchase Management application, if the Style Inventory Management application is not installed for this company.

## Purchasing Officers

If items are to be controlled by their respective Purchase Officers, then definitions must be maintained per officer.

## Item Supplier Profiles

For each supplier of an item, a profile is required within the application. These may be set up 'in flight' in order entry, or maintained as part of the initial data set up.

## Delivery Instructions

A set of delivery instructions may be maintained per company. Each set is identified by a three character code and comprises of up to three lines of text to be printed on the Purchase Order (when selected).

## Receiving Addresses

A set of receiving addresses may be maintained per company. Each set is identified by a three character code and comprises of up to five lines of address to be printed on the Purchase Order (when selected).

## Supplier External Text

For each supplier, up to 99 lines of text may be maintained, to be printed on the order when selected.

## User Authorities

For each user ID (defined to System Manager) which will be used to access the application, a value and percentage of order line value may be maintained. These are used to check in Goods Receiving and Invoice Matching if the user has sufficient authority to accept an under/over delivery, or a price greater than that specified for the order.

## Parameters And Sundry Codes

A number of Parameters must be maintained for each company. These include:

- Unit of measure codes
- Item classifications
- Order classifications
- Scrap reason codes
- Schedule order types
- Stockroom codes
- User defined fields

These are used to tailor the application to meet the company environment.

## Fax Numbers

If purchase orders are to be faxed, rather than printed, then fax numbers must be maintained for the relevant suppliers.

# Implementation Critical Path

After installing the software provided, the recommended steps for implementing Purchase Management are outlined as follows:

- Define Company Profile options.
- Create Company Profile.
- Initialise Parameter and Sundry Codes file.
- Design the units of measure table (including conversion factors).
- Define stockroom codes (for non-stock orders).
- Define order classifications.
- Define schedule order types.
- Define scrap reason codes.
- Define item classifications (where Inventory Management is not in use).
- Define user controlled fields for order entry.
- Create parameter and sundry codes.
- Define user authority values and create them.
- Define and create currency rates (unless these have already been set up in the Purchase Ledger).
- Create Purchasing Officer definitions.
- Create supplier master file (if Accounts Payable is not installed).
- Create item specifications (if Style Inventory Management is not used).
- Create item/supplier profiles. These may also be created in Order Entry.
- Define and create delivery instructions sets.
- Define and create receiving addresses.
- Optionally create extended purchase text for:
  - Items
  - Items/Suppliers
  - Suppliers
- Optionally set up Supplier Fax numbers.

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## Chapter 3

### Style Sales Order Processing

#### Overview



## Introduction

Style Sales Order Processing has a wide range functions which not only enable it to be used by a variety of industry specific organisations, but also to change as a company grows and develop in response to a changing environment. The application is designed to support both a rapid 'data entry' style of order entry for use when very few changes or overrides are required, as well as order entry carried out at the same time as a conversation with the customer.

Although the application contains a variety of pricing and discounting routines, in certain circumstances it is necessary to enter a negotiated price or discount. The order may be suspended if non-standard conditions such as zero pricing or credit check failure, arise during entry, dependant upon parameter settings. This allows order taking to continue unhindered, but errors to be trapped.

Day to day decisions are supported by a range of enquiries, which enable stock availability to be projected into the future, customer credit to be examined and orders to be displayed by customers reference, item number and account.

## Relationships To Other Applications

The Sales Order Processing application is a fully standardised Style application and, as such, operates under the control of the system manager.

The following applications are prerequisites to Style Sales Order Processing:

- System Manager
- Style Inventory Management
- Accounts Receivable

Normally Style Sales Analysis would also be installed with Sales Order Processing to provide flexible enquiry and reporting on the sales activity of the company.

# Application Configuration

As with all Style applications, Sales Order Processing can be operated for a number of companies, the characteristics of each being maintained on a control file. The application is controlled by setting up data both for the company and for each stockroom. This would normally be done after the Inventory and Accounts Receivable companies had been created. The Accounts Receivable, Inventory, and Sales Order Processing applications all use the same company codes for interfacing companies.

Certain data and policies may be set up as a default at the company level. These are:

- Company Name and Address
- Document reference numbers for orders and invoices
- Period and week numbers
- Defaults for Additional Customer details
- Nominated fields for use as search arguments for Customer Selection

These policies and parameters will be used throughout the operation of the application unless overridden at customer or order level.

When the Inventory application is set up, a number of stockrooms may be created. These will be created for a number of purposes, only some of which may be concerned with sales and distribution. Any stockroom from which sales are to be made must be designated as a 'sales order processing' stockroom and defined as such within Sales Order Processing. As well as address and contact information, allocation parameters can be maintained for each stockroom.

# Application Set Up Data

There are a number of common features in the way in which data is maintained within Style. These are:

- The standard style search routine which is used to identify styles
- The standard name and address search routine which is used to identify customers
- The standard method for selection of a code or style from a displayed list

The following reference data can be maintained:

Standard Codes and Descriptions

Customers

Styles

Prices and Discounts

## Standard Codes And Descriptions

A description or parameters file is used by Style in most applications for a number of purposes. Primarily, it enables codes to be set up with standard descriptions. This provides validation at the time of data entry and also the display of descriptions on both Screens and reports. In some specific instances a parameter or number may also be stored.

The application has a small number of standard codes and parameters that must be present for the successful operation of a specific application. As the Inventory Management application is the base application for both Distribution and Manufacturing applications, the establishment of the file is in that application. Maintenance, however, can be performed from most applications, including Sales Order Processing.

## Customers

The information held in the application for customers is of equal interest to those managing and collecting the debt as to those recording and processing sales orders. The maintenance routines in each application reflect their specific requirements. They both, however, create and maintain a common database with access to common and exclusive information. For example, the Sales Order Processing application provides functions to control stock allocation, pricing and sales analysis coding.

The structure of the customer information can be manipulated to match the structure of the customers organisation. Customer accounts can be set up as single accounts, or as statement accounts with many invoice to accounts, each of which may have many delivery accounts.

### **Statement Account**

This is shared with the Accounts Receivable application and is normally created and maintained in Sales Order Processing, if both modules are in place. The data held at this level includes credit details, Accounts Receivable parameters and statistical codes.

### **Invoice Account**

Invoices may be sent to an invoice only address.

### **Delivery Account**

Many delivery (ship to) addresses may be set up for an account. This is done by adding a three character suffix to the eight character invoice account code.

### **Additional Information**

Each account to which items are delivered may have additional data to act as defaults during order entry. This enables different parts of a Customers organisation to have different commercial terms and be serviced from a different warehouse.

## Styles

Style master information can be maintained from Sales Order Processing. This includes style description, both external and internal additional text and all of the style codes and classifications used in Sales Analysis.

Style alternatives are maintained in the Inventory application but are used in Sales Order Processing.

Styles may be created with up to 100 colours and 30 sizes and 30 fits of each. This gives a maximum of 3000 variations.

## Prices And Discounts

In order to get the full benefit from Sales Order Processing, pricing and discounting should be performed by the application. This reduces errors, and hence credit notes and customer irritation. Style provides for pricing to be performed at both order entry, using customer and style criteria, or at shipping. Alternatively, prices and discounts may be set manually.

Many price and discount lists can be created. This may be necessary for a number of reasons, for example, different market sectors, customers or multiple currencies. Discounting can be based upon quantity or value and be given in terms of percentage or value. A discount may apply to an item, a range of items or the whole order.

Effectivity dates can be used to control the application of new prices and discounts.

## Price Lists

A price list can be used throughout the company or for an individual customer. Price lists are identified by a code which is held as part of the customer additional information.

The style entries within a price list have an effectivity start date so that price changes may be entered onto the application in advance of their introduction.

Global increases by percentage can be applied to an individual price list. A new list may be created by copying an existing list.

## Discount Lists

A discount list can be applied throughout a company or to an individual customer in a similar manner to price lists - the appropriate code being set up on the customer additional details record.

Discounts can be based on quantity or value breaks, either in the form of a percentage or an amount.

A discount can be allocated to an individual item, or alternatively the same discount can be given to all items in a discount group. A discount group is an attribute of a style and is set up in Style Maintenance to group related items together for discounting purposes.

# Order Processing Cycle

Sales Order Processing has five main procedures:

- Record order details
- Stock allocation
- Pricing, discounting and credit checking
- Pick items from stock and ship
- Generate invoice

Each one of these procedures has a number of tasks each with a large number of options.

## Record Order Details

The Style Sales Order Processing Application provides two major types of order entry - transcriptional and conversational. The first is designed for batch processing and no dialogue with the customer at the time of entry. This would typically be used perhaps where orders have previously been entered onto an order pad, and a high customer service level means that as most items will be allocated, few overrides are required. Stock can never be allocated at order entry time using the transcriptional method.

To complement this, conversational order entry is designed to support a dialogue with the customer, offering alternatives in a short stock situation and agreeing on price and discount changes.

Both types of order have three phases:

- Order header
- Order lines
- Extra charges

## Order Header

The customer is identified and the standard defaults such as pricing, allocation rules, supplying warehouse and trading terms are retrieved.

## Order Lines

For transcriptional and conversational order entry, orders are entered one style at a time, with opportunity to evaluate availability and alternatives. Pricing is normally automatic, although overrides are possible. Stock allocation can be interactive with conversational order entry, but is not done for transcriptional order entry.

## Extra Charges

Standard extra charges may be held on the Descriptions file. These may be viewed and added to an order during the closing procedure. The actual charge may be retrieved from the file or entered specifically for the order. A small order charge may be set on the parameter file and automatically added to the order, should the order value fall below a predetermined value.

## Stock Allocation

When an order line is considered for allocation, whether interactively or in batch, the due date is compared with the time fence for the stockroom (this time fence is the number of days required to prepare and deliver orders to the customer). Thus stock is allocated as late as practical and maximum flexibility is retained. During batch allocation it is necessary to sequence order lines for allocation to enable high priority orders to be met. The priority of an order can be taken from the customer or entered at order time. If low priority orders are not satisfied their priority can be increased automatically. This means that at some time their priority becomes the highest and they take precedence.

The priority number is a two-digit number between 00 and 99.

Priority number 00 means that the order line always has to be manually allocated.

Priority number 98 is the highest priority number for automatic increments. Number 99 is reserved for manual entry and will force allocate.

## Pricing, Discounting And Credit Checking

In order to get the full benefit from computerised Sales Order Processing, pricing and discounting should be performed by the application. This means that it must have a fundamentally flexible approach to pricing and discounting. The actual price and discounts charged are a function of the trading relationship with the customer, therefore, the trading currency is assumed to be the transaction currency of the customers account. Similarly the price list to be used and the discounts to be given are taken from the deliver to account of the customer. This information is part of the additional data held for customers.

It is normal for the prices charged on the invoice to be dependent upon the date the order is received. There are a small number of instances where the price and discount are determined at the time of shipping. The Sales Order Processing application provides for these decisions to be made for the company in general, for the customer, or specifically for the order.

Prices and discounts are defined as follows:

### **Price**

In the transaction currency of the customer account.

Taken from a price list at order entry or shipping.

Entered and fixed at order entry.

Expressed in the selling units of the order (multiplied by a conversion factor on the price list if required). Currency can be overridden at order level

### **Discount 1**

A percentage or value.

A discount for the item as given on the discount list.

A variable discount depending upon the quantity or value of the item sold.

Or

### **Discount 2**

A discount for the item group as given on the discount list.

A variable discount for the item group depending upon the A percentage or value.

A discount for the whole order as given on the discount list.

A variable discount depending upon the quantity or value of the whole order, quantity or value of the item sold.

### **Credit Checking**

Credit checking is performed at the following times within the application:

Order amend - providing the parameter is set on the company parameter file.

Picking - providing the allocation credit check flag on the company profile is set on.

Different types of credit check can be applied within the application, depending upon the setting of the extended credit check parameter and grace period parameter on

the company parameter file. If the extended credit check parameter is set to zero (No), then the application assumes a credit type of 1 will always apply. If the extended credit check parameter is set to 1 (Yes), then the following credit types may apply:

- 0 - No credit check
- 1 - Credit limit only
- 2 - Credit limit & debt outside of payment terms
- 3 - Debt outside of payment terms

If a number of grace days are entered as a parameter, then these days are added to the normal debtor days before performing the credit check. The actual number of days is set on the Descriptions File using type GRAC.

The application provides for the storage of two credit limits -commercial and insured. The commercial value is used for credit checking purposes.

Orders for customers who have exceeded their credit limit are suspended with the code CL, and for customers who have been placed on credit stop with the code CS.

## Picking And Shipping

Selection is provided to ensure that only the picking documents required are actually printed. This selection can be for a number of days, a range of order numbers or a particular warehouse. This allows stock to be allocated to an order well in advance of shipping, but the documents to be printed only when action is required.

This is particularly useful in dealing with large export consignments. When a picking note is printed, all allocations, irrespective of when they were made, are included. This stops documents being produced unnecessarily. If the Warehousing application is installed, requests to ship are released to the warehouse in a similar manner. It is then possible to print consolidated picking lists sequenced by location.

When shipping has taken place this is confirmed to the application. A number of options and facilities are provided.

- A number of shipments may be confirmed together provided that the shipping method and the ship date are common and there are no changes
- An order can be confirmed on its own. This enables additional charges to be added
- An individual shipment may be displayed, amended where necessary and then confirmed

# Invoicing

There are a number of ways to request an invoice. These are:

- Confirm the shipment of an order
- Request a miscellaneous invoice - with stock update
- Request a miscellaneous invoice - without stock update

There are also requests for credit notes that operate in an identical fashion to the requests for miscellaneous invoices.

When an invoice or credit note is generated in the application, prices and discounts are either taken from the order, as in the case of a confirmed shipment or created during entry using standard rules. There is a special facility to produce a credit note that is based on an invoice or a stock return. This can be used to reverse a previous posting that was perhaps raised in error.

Having generated the invoice or credit note a procedure is run to post to Accounts Receivable, the General Ledger, and to print documents. Invoices and Credit notes are printed as two separate documents.

An invoice will be raised and printed in the transaction currency of the customers account. Postings will be in both transaction currency and the base currency of the Accounts Receivable. The data that is passed to the Sales Analysis application will be in the base currency of the company.

# Enquiries

A significant part of the work done in a busy sales office is answering questions. These are typically regarding:

- The status of an order
- The prices being charged and discounts given
- The date on which a delivery can be expected
- Invoice or credit note queries

Style provides a comprehensive range of enquiries to view:

Orders by :

Order

Customer

Style

Cross Reference (customer reference)

Invoices by:

Invoice

Order

Customer

Stock availability - Time-phased expected stock position.

Stock position - Stock holding at each warehouse.

As well as these day-to-day enquires, the following application reference data can be displayed:

- Customers
- Styles
- Price Lists
- Discount Lists

# Reports

Reports can be grouped into reference data, status, exception and action reports.

Reference data :

Customers

Styles

Price Lists

Discount Lists

Commission Lists

Status

Outstanding orders

Back orders

Stock versus outstanding orders

List of orders

Orders summary

Forward orders

Rolling sales forecast

Exception

Unconfirmed picking notes

Suspended orders

Orders with zero price

Gross margin exceptions

Selling price overrides

Zero value invoice lines

Action

Consolidated Pick notes

Pick Lists

Special Tasking report

## Implementation Considerations

All Style documentation emphasises the vital importance of careful and detailed planning prior to the introduction of the new application. When Sales Order Processing is installed, two major areas must be considered. These are:

- Have the pre-requisite applications already been installed; and if so, are they set up in a way which is suitable for the operation of Sales Order Processing ?
- How will the functions and facilities of the Sales Order Processing application be used to satisfy the companies basic needs?

This section deals with each issue in turn.

## Relationship To Other Applications

The fundamental purpose of the Sales Order Processing application is to record and control your sales order. These orders are for the products of the company and therefore there is a direct relationship between the Inventory Management application and this application. One Sales Order Processing company must be set up for each Inventory company for which sales are to be managed. Similarly, Sales Order Processing will raise invoices for all despatches made from the company and it will expect to post them to the customers accounts in Accounts Receivable. There is, therefore, a simple relationship between the applications for one company; that is, the applications of Inventory, Accounts Receivable, Sales Analysis and Sales Order Processing must share the same two-character Company Code.

A number of Sales Order Processing functions depend directly on data held in the Accounts Receivable and Inventory Management applications. These are:

- The currency of the Sales Order Processing application is determined by the currency setting of the Accounts Receivable company
- The currency of a customer is determined by the currency setting on the customers Accounts Receivable account
- Only alternatives set up in the Inventory Management company will be prompted as replacements for shortages during sales order entry
- The Analysis Codes on the Item Master record are not only used within the Inventory Management application for reporting, but also form the basis of Sales Analysis. Considerable care must be taken to ensure that the coding structure which is set up will meet the analysis needs for the foreseeable future, as the regeneration of the Sales Analysis database can take a very long time

## Sales Order Processing Utilisation

When the applications goes 'live', sales orders must be recorded accurately and quickly. Prices and delivery dates are quoted to customers, and your companies reputation is exposed.

It is essential, therefore, that detailed discussions take place which review the facilities offered by the application and decisions made as to how each stage of the order cycle will be handled. For example:

- In what circumstances will conversational order entry be used?
- Will manual pick lists be raised?
- Under what circumstances will order acknowledgements be sent?
- What use will be made of the priority allocation rules?

All of the above are fully discussed in the body of this manual. Some of the decisions require control data to be set up in the applications in order to implement them; others are a matter of agreed procedure; all directly affect the service which you provide to your customers.

## Basic Data Set Up

Having decided how you will use Sales Order Processing in conjunction with other applications and the companies you will use, you need to plan and implement the setting up of the basic data required for the applications operation. The main elements you will need to create are:

- Company profiles
- Warehouse profiles
- Price & Discount lists
- Customer Details
- Descriptions
- Items
- Alternatives

The maintenance of these elements is covered in detail in the Application Set up Data section of this guide. It is sufficient here to highlight particular issues to bear in mind when planning the implementation.

## Company Profile

The issues which you should address prior to setting up company profiles are:

- Will you wish generally to allow back orders and partial deliveries?
- What is the default priority which you wish to give to new orders, and what are the steps by which that priority is to be incremented when allocation fails?
- Do you want to allow allocation to a level where available stock is negative?
- Do you, as a rule, wish to price orders at order, or despatch time?
- Which fields in the customer database do you wish to form part of your Customer Search facility

## Warehouse Profiles

As part of the implementation of the Inventory Management application, you will have set up one or several stockrooms. If any of these stockrooms are to be used to process sales orders, Warehouse Profiles will have to be set up for them in the Sales Order Processing company. For each warehouse you will need to define in particular the normal allocation priority and allocation time-fence for sales orders at this warehouse.

## Descriptions

The Sales Order Processing application holds a file of parameter codes and descriptions which is used by itself and other applications. The application makes use of this file. Many description codes will have been set up as part of the implementation of Inventory Management, but there are codes specific to Sales Order Processing which you will need to set up, such as customer classes, cash discounts, extra charges, etc.

## Price And Discount Lists

The Sales Order Processing application enables multiple price and discount lists to be maintained. Points to consider prior to setting up these lists include:

- Who, within your organisation, will be allowed to maintain lists? This authority is specified on the price/discount list profile which must be set up for each list
- How many lists do you require? Will you wish to set up lists which are unique to individual customers, or for use on special orders?
- Within your discount lists, will you be able to specify discounts for groups of items, or are they to be specified for individual items? Will you use whole order discounts?

## Customers

The main statement account details for your customers may have been set up as part of the implementation of Accounts Receivable. Specifically for Sales Order Processing you will need to set up:

- Invoice and delivery address details -- this application enables the set up of many addresses for invoice and delivery to points for a customer
- Default parameters for controlling the way in which orders for customers are processed, such as the default sales, warehouse, price list, discount list, allocation priority and parameters controlling back orders and partial deliveries
- The customer analysis coding, such as division, territory, salesman, customer class, etc.

## Items

Item descriptions and basic data will have been set up as part of the implementation of Inventory Management. It may be worth while reviewing the following, as part of the implementation of Sales Order Processing:

- Analysis coding, i.e. item group, item class, etc.
- Item text - this may be printed in external sales documents

## Alternative Items

Alternative items are used within the Sales Order Processing application, but are set up and maintained in Inventory Management. If you do set up alternatives, it is important to decide whether a relationship is reversible.

## Initial Data Take On

Once all the basic data has been set up, you are ready to use the application to process sales orders. You may wish to convert outstanding sales orders from another application - this may be done by specially-written programs. Alternatively, you may decide to continue to process existing orders on your old application, and enter new orders, from a given date, into the new application. By this method you may gradually run down orders on the old application and build the order bank on the new, but great care must be exercised over the updating of the inventory figures.

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## Chapter 4

# Style Inventory Management Overview



# Introduction

The world of commerce can be divided into organisations which provide goods and those which provide services. All of those concerned with the acquisition and supply of goods are required to record, count, manage and account for inventory. It is one of the fundamental needs of all such companies. A system to keep proper control of inventory is a prerequisite for buying decisions, answering customer questions and deciding the production program. Different organisations require different levels of detail and support for Inventory Management decisions, and indeed this might vary within the same organisation. However, the basics of defining items, recording movement, even calculating balances and providing valuations will be common to all such companies.

The Style Inventory Management application provides a wide spectrum of facilities for the recording, valuation and management of inventory. These facilities are the bed-rock on which the interface to the Style product are built. The application may be installed in a simple organisation which views its inventory as a single logical unit, or with equal ease in a multi-warehousing organisation with separate raw material, finished goods and component stores. Multiple stockrooms or stores may be created, with the ability to maintain different stock investment policies in each. This is achieved without data redundancy as the same style definition is used throughout.

## Relationships To Other Applications

The Style Inventory Management application is a fully standardised application and as such operates under the control of the System Manager.

There are no pre-requisite applications for this module (apart from the System Manager), moreover this application is a pre-requisite for all the interfaced applications.

Although the application is designed to operate as a fully integrated application of Style, it can be operated in a very simple fashion as a stand-alone system, including facilities to record Sales and Purchase activity as well as providing Inventory recording and control.

# Application Configuration

As with all Style applications Inventory Management can be operated for a number of companies, the characteristics of each being maintained on a control file. The application is controlled by setting up data for the company and then for each stockroom within the company. Stockrooms may be added after the initial implementation of the application.

Certain policies may be set up as a default at the company level. These are:

- Costing Method
- General Ledger codes
- Period for usage calculation (week or month)
- Stock Management Policy factors:
  - Method of calculation
  - Re-order quantity factor
  - Stock carrying rate
  - Supply order placement cost

These policies and parameters will be used throughout the operation of the application unless overridden at the level of the item or at the time of the transaction.

It is necessary to define each of the stockrooms where stock will be held. Three important policies can be set at this level. First, the stockroom may be excluded from valuation, i.e. for consignment stocks. Second, certain transactions may be excluded from use in the stockroom. Finally a value usage matrix can be created to manage the stock investment policy of the stockroom. Nine value/usage bands may be created and target minimum and maximum weeks of stock defined. In order to calculate average usage, two factors must be defined: (1) the time interval in weeks or months and (2) the periods to be taken into account in the calculation of average usage. The time interval is set on the company record and a number of profiles may be maintained detailing which of the previous sixty time periods (weeks or months) will be included in the calculation.

# Maintenance

The maintenance functions of the Style Inventory Management application enable the creation and amendment of master files and data files which are fundamental to its operation. The application will then build the necessary transaction and movement histories as work progresses.

There are a number of common features in the way in which data is maintained, these are:

- The standard style (product) and description search routines
- A standard method of selecting codes from a displayed list

The following data can be maintained:

- Standard Codes and Descriptions
- Style (product) Master Details
- Alternative Products
- Stockroom or Balance Details
- Calendar

## Standard Code And Descriptions

A Descriptions or Parameters File is used by Style in most applications for a number of purposes. Primarily, it enables codes to be set up with standard descriptions. This provides for validation at the time of data entry and also the display of descriptions on both screens and reports. In some specific instances a rate is also stored, for example, in the case of the VAT Table. The application has a small number of standard parameters that must be present for its successful operation. The Style Inventory Management Application Descriptions file is the one used by Style applications. The standard parameters held on this file, therefore, include those required by all Style applications.

## Style (Product) Data

Style Inventory Management is designed to have a flexible data structure, but without incurring data redundancy. In order to achieve this there is a home Style (Product) Master record which identifies the product to the application and indeed to all applications. This holds the static descriptive and classification data. If more data is required in Inventory or any other application then a record of that information is set up and maintained in the originating application.

The following are examples of the additional data:

### Inventory

- Stockroom balance records
- Lot or batch records
- Extended internal and external text
- Alternatives
- Search family codes

### Sales Order Processing

- Pricing Details
- Default Size Distribution

### Purchase Management

- Item/supplier profiles
- Manufacturing Applications
- Manufacturing profile

### Forecasting

- Style Product/Forecast profile

### DRP

- Style Product/DRP profile

In this way it can be seen that Style requires the entry and maintenance of only that data necessary to run the operational applications in any company's system.

The Style Product Master details can be grouped into the Static Part record which includes:

- Description
- Six different Analysis Codes
- Units of Measure - three may be defined for each Style (Product)
- VAT Codes
- Costing method, costs and default General Ledger Codes

- Sourcing Warehouse (Primary Warehouse)
- Supersession details
- Search Families

## Stockroom Balance/Warehouse Select

For any Style/Product one or more stockrooms may be defined. A stockroom is a discrete area within a company where stock is recorded and controlled separately from other company stocks. Depending upon the type of company, a stockroom may be used to define divisions, depots, warehouses, stores, an area within a store or even van stock. If such distinctions are not required a single stockroom is created to record the total company stock of the item.

The Static Warehouse information is concerned with:

- Units of Measure at the Stockroom
- Inventory Management Policy and Control
- Balance Data
- Costs
- Preferred Suppliers

The warehouse select allows the user to select more than one warehouse where the style/product may be stored, etc.

## Batch Control / Lot Traceability

Batch or Lot tracking is available in style. If warehouse control is required, the Warehousing application should be installed.

## Alternative Styles

For any style one or more alternatives may be nominated. When the relationship is created it is possible to state whether or not the relationship is reversed. This facility is used by the Sales Order Processing functions to offer customers alternatives if a style is not currently available.

## Calendar Control

The Inventory Management application deals with dates, week endings and period endings, in order to perform reconciliation, usage calculations and information analysis. Both week end dates and period end dates can be maintained for all operational years.

## Colour File

This holds the selection of colours which are available across all the styles that are stocked (this can total 100).

## Size File

This is a selection of sizes available across all the styles that are to be stocked. The size file is linked to a Size Mask file.

## Size Masks

A size mask is a grouping of sizes that apply to a particular style range. Sizes can be included in more than one size mask, if required. Two size masks are used when a four dimensional product is designated a style range.

## Cartons

Cartons are a collection of piece items that can be comprised of various colour/sizes/fittings all within the same style code.

## What Is Carton Processing?

Carton processing allows a group of items to be processed as a single item. The group is a carton containing a mix of colours and sizes (and fitting, if four dimensional) for a particular style:



Red Small



Red Medium



Red Large

The Carton of Suitcases contains:

5 x Red Small + 3 x Red Medium + 2 x Red Large

If the product style is four dimensional (it has a fitting) this would be dealt with in the same way.

This is of use to companies who import from the Far East where goods are commonly packed in set ratios. The importer commonly wants to despatch these goods without breaking open the carton but needs to know what is inside the carton.

## Key Features

- Carton set-up allows the ratio of colours, sizes (and fitting) within a carton to be entered.
- Carton enquiry shows stock in cartons and pieces.
- Carton breakdown may be printed on any external documents where items are involved.
- Sales analysis may be updated with cartons sold or pieces sold.
- Carton stock adjustment allows a carton to be broken open.



Tip: This results in the stock of pieces being increased and the stock of cartons decreased.

## Using Cartons

To use carton processing, a carton profile for each company must be defined. This controls the carton function within each company, allowing different companies to use cartons in a way which suits their particular needs.

To set-up cartons, two separate styles must be defined:

- A style to represent the carton, this would typically have colours and may be a size/fitting element that reflected the number of individual pieces within a particular carton.
- A style to represent the pieces within the carton. The pieces would be set-up with colours, sizes (and fitting if required).

Once these two styles have been defined the carton may be built using carton maintenance. Here, a carton is defined in terms of the pieces within the carton.

Processing is at carton level - orders placed for cartons and stock are then held at carton level. If a carton is opened (for example a sample may be required) the carton stock adjustment facility may be used. This will down date the carton stock figure and increase the stock of pieces according to the ratio of pieces in the carton.

The carton breakdown in terms of pieces, is displayed in the carton enquiry and on external documents such as invoices which are sent to customers.

# Processing

There are four main processing activities involved in the operation of the Inventory Management application. These are as follows:

- Movement Transactions
- Stock Counting
- Stock Management
- Periodic Processing

The first two functions are concerned with the performance of the stock recording aspects of Inventory, the third is concerned with deciding what stock should be where, and the fourth with building the database to aid management decisions. As the movement transactions are applied to the balances interactively, the displayed balance reflects the latest information in the system. Therefore, supply and despatch decisions can be made in greater confidence. The Inventory application can be installed in isolation. However, if Purchasing, Manufacturing and Sales Systems are installed most of the movements will be performed in these operational systems. The Inventory application will concern itself exclusively with counting, housekeeping and stock management decisions.

## Movement Transactions

The primary purpose of an Inventory movement is to record an action within the application and to calculate its effect on the stock balance. In its simplest form a miscellaneous receipt booking straight into Inventory increases the physical stock. However, a Customer Order Issue, will not only reduce the physical stock but it also reduces the balance of the stock allocated to orders.

The following movement types are available:

Customer Order Issue

Miscellaneous Issue

Purchase Order Receipt

Miscellaneous Receipt

Adjustment to Physical Stock

Adjustment to Allocated Stock

Adjustment to Frozen Stock

Transfer Out to Another Stockroom

Transfer In from Another Stockroom

Record Purchase Order

Transfer In and Out (Move from one stockroom to another on a single transaction)

Carton Stock Adjustment

Each movement updates the appropriate balance field on the style/product stockroom balance. If the style is defined as being Batch/Lot or Serial controlled the additional control number must be entered. A subset of the balances are maintained for the lower level record.

Costs may be entered with the movement and depending upon the movement type and the costing method the database will be updated appropriately.

There are four ways of valuing stock movements. These are:

### **Average Cost**

The average cost of stock in the stockroom is recalculated everytime a receipt is made. This average cost is used to value issues. (Receipts are valued at receipt cost).

### **Standard Cost**

The standard cost for the part is either 'rolled over' from the manufacturing application or directly maintained. This cost is used to value all movements, both receipts and issues.

### **FIFO Cost**

This is the acronym for **F**irst **I**n **F**irst **O**ut. Each receipt for the Product is maintained as a separate balance and as an issue is made the product is consumed. The first receipt to arrive is the first receipt balance to be used for issues. The cost of the

issue is taken from the receipt balance. The FIFO sequence is maintained by time as well as date. If the part is lot controlled then the FIFO operates **within** lot.

### **Latest Cost**

This is when the latest cost of the receipt issued is used to value the stock.

The default or preferred costing method is set on the company control record. However, you can control at the level of the product when it is defined and set a different costing method for that particular item. All costs are kept in up to five decimal places of base currency. The latest cost of all the product is recorded and the average cost calculated whatever the costing method. If, for example, the valuation method is FIFO this can be simply compared with the average and latest cost on an enquiry screen.

For each inventory movement a record is written to a transaction file. This provides a full audit trail of movements. The information held includes:

- Movement type
- Supplier or Customer
- Quantity moved
- Value of the movement
- Resulting stockroom balance
- Batch/Lot number
- Date and Time of movement
- Person responsible
- Terminal used

The movement week and period details are determined from the calendar and period files, and set to either the current or next week/period.

The movement history may be held on the system as long as necessary but not less than one month. This is because it is used to prove the closing balance at the end of the period. It is available for both Enquiry and Reporting to assist investigations into stock inaccuracies and provide the basis of activity analysis.

## Stock Counting

The application provides a simple to use stock counting procedure which incorporates cycle counting. This enables the automatic creation of count recommendations. the following steps are involved:

- Count creation - this is done manually by item ranges, or automatically on a time based cycle.
- Review and maintain count recommendations.
- Request printing of Stock Count sheets
- Enter counted Stock figures
- Reconcile Stock Count
- Post adjustments if applicable

Products are selected for stocktake by any of the following options:

- Product Number
- Product Type
- Product Class
- Product Group Major
- Product Group Minor
- Stockroom

The application records the stock level of the selected product(s) at the start of the count. This enables everyday processing of movements to continue while stock is being counted and reconciliation to take place on completion of the count. Stocktake count sheets are then produced to the required detail, i.e. all levels of inventory records are exploded for the selected items. These may be used as input documents to the following processes.

The total quantity of physical stock of each product counted is entered from the stocktake sheets. This data is validated and a report produced.

Until the stocktake is deemed complete, a reconciliation report of physical stock versus computer stock count is available. This would normally be selected by the same parameters as used when requesting the original stock count sheets. The report highlights the differences between physical and computer figures with actual quantities and value variance. This is to help assess the importance of any discrepancy. On completion of the stocktake the application offers the option to generate automatic adjustment of any unresolved discrepancies. Any variances outside the tolerances defined on the stockroom count profile are designated as count errors.



Tip: Once the count sheets have been requested, no stock movements should take place for the selected products until after the physical counts have been entered and reconciled.

To assist in the reconciliation of any discrepancies a report is available giving summary or detail information of stock movements.

## Stock Management

The system offers a number of stock management techniques including Material Requirements Planning and Distribution Requirements Planning. The Inventory Management application, even in isolation, can be used to operate an order point stock management policy. The main factors in this policy are:

- Average Usage
- Reorder point and maximum stock
- Reordering rules

## Average Usage

The usage of an item is held on the stock history file and is maintained either weekly or monthly according to the usage cycle specified on the company profile record. Average usage is re-calculated during week or month end processing as appropriate.

The usage profile defines which of the preceding weeks or months usage are to be included in the calculation of average usage. The product stockroom balance record holds the code for the usage profile which will be used in the calculation of average usage.

Typically usage profiles will be set up to represent:

- Three months moving average
- Twelve months moving average
- Seasonal moving average (This would be achieved by taking the previous month and the 'next' month last year)



*Note: Care is taken to ensure that the average is only calculated from the start of the product's life even though the profile indicated that previous data should be included. This is to ensure that the average is not distorted by the absence of history data.*

## Reorder Point And Maximum Stock

The parameters of maximum stock, reorder point, reorder quantity and economic order quantity may be maintained for each product in each stockroom. Alternatively, by using the historical usage information, maintained as a result of all movements, the application can calculate these figures. These calculations are based upon grouping products into the categories defined on the stock weeks matrix:

Nine categories are used:

|   |              |              |
|---|--------------|--------------|
| 1 | Low value    | Low usage    |
| 2 | Low value    | Medium usage |
| 3 | Low value    | High usage   |
| 4 | Medium value | Low usage    |
| 5 | Medium value | Medium usage |
| 6 | Medium value | High usage   |
| 7 | High value   | Low usage    |
| 8 | High value   | Medium usage |
| 9 | High value   | High usage   |

The definition of these groups is held on the stock weeks code matrix which is updated as part of the Company Profile Maintenance procedure.

Limits are set to define the boundaries for high, medium and low usage and value. Minimum and maximum stock holdings, expressed as weeks at average usage, are assigned to each of the nine categories. The minimum stockholding in weeks is the number of weeks stock that will be left after the lead time has expired, and therefore, represents a safety stockholding. Safety stock is a measure of how likely the lead time and average usage are going to increase.

When the procedure to calculate Reorder Point, Maximum stock, etc., is run, the application reads the stockroom records and re-assesses, for allowed products what the reorder point and maximum stock quantities should be.

The application assesses where the product falls on the value axis of the stock weeks matrix based on its cost. It then determines where it falls on the usage axis based on its average usage.

Having assessed which of the nine categories this product falls into, the minimum and maximum stockholding (in weeks) are taken from the matrix and the following calculations performed:

Reorder Point=(Lead Time + Minimum stockholding) × Average Usage

(Lead time and Minimum stockholding are expressed in weeks)

Maximum Stock=(Lead Time + Maximum stockholding) × Average Usage

(Lead time and Maximum stockholding are expressed in weeks)

Reorder point and Maximum stock are held on the product stockroom balance record.

Since average usage is used in the calculation procedure, it is best run after the week end or month end processing procedure in which average usage is re-calculated.

The application is also able to calculate an Economic Order Quantity as follows:

$$EOQ = \sqrt{\frac{2 \times AU Y \times OPC}{IC \times SCR \times 100}}$$

where:

|     |                           |
|-----|---------------------------|
| EOQ | = Economic Order Quantity |
| AUY | = Average Usage Per Year  |
| OPC | = Order Placement Cost    |
| IC  | = Item Cost               |
| SCR | = Stock Carrying Rate %   |

## Reordering Rules

The application supports five different reordering policies:

### **Up To MAX**

When expected stock drops below the reorder point the system suggests that an order be placed of sufficient size to take the stock level up to the maximum.

### **Reorder Quantity**

When expected stock drops below the reorder point the application suggests that the reorder quantity be ordered. This quantity is either set by the user or determined by the application. A Reorder Quantity Factor can be held at stockroom level or a default is held on the Company Profile. If the reorder quantity factor was 30%, the application will recommend action to reorder enough stock to last through 30% of the lead time. If the factor was 200% the application will recommend action to reorder enough stock to last through twice the lead time.

### **Up To ROP**

When expected stock drops below the reorder point the application suggests that enough stock should be ordered to take the level up to the reorder point again. For example, if the reorder point was set to zero, this rule is the one that would ensure that goods were only reordered when there is demand for them in the form of a customer or manufacturing order.

### **Up To Min**

When expected stock drops below the reorder point the system suggests that an order be placed of sufficient size, to take the stock held up to the minimum level.

### **EOQ**

When expected stock drops below the reorder point the system suggests that an order be placed for the calculated or maintained economic order quantity.

## Available/Expected Stock

These are calculated as follows:

Available Stock=Physical Stock - Frozen Stock - Allocated Stock - Picked stock

Expected Stock=Available Stock + Stock In Transit + Stock On Supply Order - Stock  
On Back Order

## Periodic Processing

Although the Inventory management application is a transaction based system which records events as they happen, many of the decisions concerning stock levels and raising and amending orders are based upon the performance of a product over time. In order to make these decisions it is necessary to be able to view previous periods' activity. This is achieved by maintaining history files, not only of the transactions, but also summarised by type of transaction. These are maintained by running weekly, monthly and year end procedures.

# Enquiries

The enquiries in the Inventory Management application allow a multi-layer view of a style / product. These layers are:

- Company stock summarised by stockroom
- Full details of the current balance, usage, and cost at a stockroom
- The breakdown of a stockroom balance into the batches or lots that comprise the stock holding
- The movements that gave rise to the present balance
- The summary of activity for previous periods
- In addition, the reference files are supported by a full range of enquiries.

# Reports

The reports produced by Inventory Management can be classified as:

- Status
- Audit
- Valuation
- Movement Analysis
- Action

Status reports are provided on the master files and particularly on the style file. Audits with before and after images are printed on request. These include all changes since the previous print.

Most of the reports are supported by both selection and sequence parameters. This helps to minimise the processing time and the amount of paper required.

## Implementation Considerations

It is of vital importance to the success of any software implementation that some fundamental issues of the implementation are considered before any detailed steps are undertaken. In the case of Style Inventory Management these issues are as follows:

- How do you wish to use Style Inventory Management and other Style applications?
- How many Inventory Management companies do you intend to set up?

These issues are usually closely linked, and this section deals with each issue in turn.

## Inventory Management Utilisation

A fundamental point to consider here is what functions do you intend Inventory Management to fulfil? Are you intending to use Inventory Management on a stand-alone basis, and to record all stock movements through its own procedures? If this is the case then this obviously simplifies the implementation, as you only need to be concerned with the set up requirements for Inventory Management.

If, however, you are installing other Style applications, then your Inventory Management implementation needs to take into account the requirements of those applications. As Inventory Management is a pre-requisite for all the Distribution and Production applications of the system, you can appreciate the need to consider its use in relation to the other applications.

Where Inventory Management is installed with other business applications it provides:

- The basic database definition of parts/styles, stockrooms and code descriptions
- The central repository for information on stock holding, whether at stockroom or lot level
- The central audit of stock movements and history
- The source of costing information for the valuation of movements and stock holding
- The other Style applications, therefore, use Inventory Management as shown in the following examples:
- Purchase Order Management uses the Inventory part definition; holds 'on-order' information as a stockroom balance quantity; and posts movement transactions to Inventory Management for receipts and returns
- Sales Order Processing & Invoicing uses the Inventory style/product definitions; makes use of parameters defined on the descriptions file; holds back order and allocation quantities as stockroom balance fields; and posts movement transactions to Inventory Management for customer issues and returns
- The Production applications use the Inventory basic part/style definitions; hold 'on-order' and allocation quantities as stockroom balance files; and post movement transactions to Inventory Management for work order issues and receipts

In a full Style implementation, it can, therefore, be seen that Inventory Management provides both basic data definitions and a transaction repository for a number of other applications. This is, of course, in addition to the functions which are only available within Inventory - i.e. the Stock Management and Stock Taking facilities, and the Stock Adjustment Transactions.

When planning your implementation it is, therefore, important that you consider:

- Which other Style applications do you plan to install?
- What requirements, in terms of data, will these applications look to Inventory Management to fulfil?

- How will the various stock movement requirements of the business be satisfied - by Inventory Management, or by one of the other applications?
- Will any non-Style applications require an interface to Inventory Management?

It is important that these sorts of questions are addressed at the planning stage. Inventory Management has a central role in any implementation, and subsequent changes are, therefore, difficult to make without disrupting the operation of other applications. It is worthwhile spending time early in the implementation to plan your use of Inventory Management and its relationship with other applications.

## Inventory Management Companies

Inventory Management, like all other Style applications, allows the simultaneous operation of discrete companies. Multiple companies are used for many reasons, but the major uses are either organisational, or for testing purposes.

A decision to set up multiple Inventory Companies for organisational reasons may very well be taken because you require separate inventories. It is more likely, however, that multiple Inventory Companies are required because you have separate operational organisations. For example, you may require separate selling organisations to be reflected in separate 'Companies' in the Sales Order Processing and Invoicing application. In a standard installation, this would require separate Inventory Management Companies to be created. Inventory Management Companies interface on a one-to-one basis with Companies in Purchase Management, Requisitioning, Sales Order Processing, Sales Analysis, Telesales, Distribution Requirements Planning and the Production applications.

It can, therefore, be seen that any decision about the use of Companies in Inventory Management needs to be taken in terms of the broader implementation.

## Basic Data Set Up

Having decided on how you will use Inventory Management in conjunction with other applications and the Companies you will use, you need to plan and implement the setting up of the basic data required for the application's operation. The main elements you will need to create are:

- Company profiles
- Stockrooms
- Stock weeks matrices
- Usage Profiles
- Descriptions
- Style/Products/Colour/Size/Fitting/Size Masks
- Stockroom Details
- Alternatives
- Calendars

The maintenance of these elements is covered in detail in the Application Reference Data section of this manual. It is sufficient here to highlight particular issues to bear in mind when planning the implementation.

Two points which may ease the actual setting up of the data are:

- The use of the Copy Company utility - this is the function which can be used to set up new Companies within Inventory Management. The essential description codes are set up automatically as part of this routine, if a company is created from the company shipped with the application.
- You may choose to convert some of the volume elements, such as parts/style and Stockroom Details, from an existing system. If you have high volumes this is particularly useful and will obviously save a significant amount of data entry time.

## Company Profile

The issues you should address prior to setting up company profiles are:

- Which costing method do you intend to adopt as a company standard? (Standard, Latest, Average and FIFO are available)
- How do you intend to make use of the user-defined specifications?
- Do you want average usage calculated on a weekly or monthly basis?
- What is to be your default reorder policy?



Tip: When initially creating the Inventory Management company profile use the Copy Company option from Utilities. This copies the Descriptions Major types and codes. Copy from the Z1 company initially.

## Stockrooms And Stock Weeks Matrices

You will need to decide how many stockrooms you require. Remember that these may be physical stockrooms or they may be 'logical' stockrooms, for example Quarantine or Written-off stock. For each stockroom you will need to define:

- Whether the stockroom is to be included in stock valuation
- Which transactions are allowed
- If the stock management facilities of the application are to be used, you will need to define the stock weeks matrix for the stockroom. Therefore, you will define the parameters of value and usage for the nine item categories, along with the preferred stock levels for each category

## Usage Profiles

The application will automatically calculate average usage based on user-defined profiles. You will need to decide how many of these profiles you need to reflect the different patterns of demand in your business.

## Descriptions

The Inventory Management application holds a file of parameter codes and descriptions which is used by itself and other Style applications. These codes cover such elements as product groups, product types, units of measure, reason codes, etc. The major codes, i.e. the subject headings, are set up automatically if you use the copy company utility to create your company data initially. Before any further data set up can take place you need to decide on the codes which you will require for each of these major headings.

## Styles/Parts

There are a number of issues to resolve prior to setting up your basic part/style definitions:

- Prior to setting up styles-colours, sizes, fittings, size masks and stockrooms must be set up.
- How are the style/parts to be coded? It is generally recommended that the style/part number should merely be a unique code to identify the style/part - not a meaningful or structured code.
- How will you use the major analysis codes which can be associated with a style/part - product major group, product minor group, product type and product class? These codes are important, as many reports allow selection and sequencing by them.
- What are the valid units of measure that you will require?
- Will any parts be subject to a costing method which is not the company standard?
- Which General Ledger accounts do you wish to associate with product/style for General Ledger updates from within style?
- What level of control do you wish to exercise over individual products/styles? Do you require lot/batch control, or is part/stockroom the lowest level you require?

## Stockroom Details

You will need to set up a stockroom details definition for each product/style which you wish to hold in a stockroom. The issues to be decided here are:

- What units do you wish the stock balances and unit costs to be expressed in?
- What are the rules regarding the preferred stock level of this product/style in this stockroom? Are they to be set automatically, or manually? Which re-order policy and usage profile are to be used?
- What is the standard cost?
- Who is your preferred supplier?

## Alternative Products

Alternative products are used within the Sales Order Processing & Invoicing application, but are set up and maintained in Inventory Management.



Tip: If you do set up alternatives, it is important to decide whether a relationship is reversible.

## Calendars

Immediately prior to processing you will need to set up the current year, period and week details via calendar maintenance. You must also ensure that week and period end dates have been set up for the current year.



*Note: It should be noted that before processing begins for a new year you need to ensure that the end dates are set up.*

## Initial Stock Balance Take On

Once all the basic data has been set up, you are then ready to appoint the initial stock balances prior to processing. This task may be undertaken as part of a conversion from another system, but where this is not the case the following inventory movements can be used:

- Physical Stock - use the Miscellaneous Receipt transaction to set the initial shelf stock
- Allocated Stock - use the Allocation Adjustment transaction to set any existing allocated stock level
- Frozen Stock - use the Frozen Stock Adjustment transaction if you currently have stock which is unavailable for issue
- Purchase Order - use the Record Purchase Order transaction to reflect the current supply situation

It is worthwhile entering these movements with a narrative indicating 'INITIAL SET UP', so that they may be readily identified in future. Making use of a standard procedure for this exercise means that a full audit is maintained in terms of movement records.

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## Chapter 5

### Style Advanced Shipping Notices

### Overview



# Introduction

Style Advanced Shipping provides the following functions, which are described more fully later:

- Master File Maintenance.
- Build and Maintain Shipping Consignments.
- Stock Reservation.
- ASNs and ANSI X12 856 ASNs.

Style Advanced Shipping Notices is an advanced business module and this overview assumes that you are familiar with the following:

- A knowledge of how the Server operates.
- A knowledge of how the JBA modules work.
- An understanding of Customer Scheduling in a repetitive manufacturing environment.
- Knowledge of how goods are collated and despatched.

# Master File Maintenance

The Style Advanced Shipping Notices module aims to cater for the special requirements of repetitive supply chain industries, for example in the apparel or footwear sectors.

The module allows you to manage and monitor complex supply contracts. Style Advanced Shipping Notices provides maintenance functions which are explained in more detail later in this section:

- Company
- Customer
- Contract

These are repetitive functions which filter the default values down from company to customer, and from customer to contract level. This minimises the amount of keying involved in setting up a new contract whilst allowing the flexibility of overrides at the lower levels.

Advanced Customer Scheduling calls Advanced Shipping Maintenance Default screens when you enter new contracts and customers. These defaults are also maintained within Advanced Shipping Notices. Other key Advanced Shipping Notice master files are:

- Name and Address. These define the EDI 'Trading Partner' details for sending Advanced Shipping Notices (ASN's) to be sent by EDI.
- Pack and Pallet Information. These define the parameters of a particular pack type; a particular pallet type and the relationship of a customer / item with a specific pack and pallet.

## Build and Maintain Consignments

This function allows you to create or maintain consignments on which to ship to a specified customer on a specified day. It is the main area of the Advanced Shipping Notices processing function.

- Default action quantities are suggested; sourcing is suggested against the available stock in priority sequence until either the requirement is satisfied or no stock remains. Suggested stock allocations are in pack rounded quantities where appropriate. Action quantities can be forced through which will make available stock negative. This is assuming the stockroom has been set up in Style Inventory Management to accept negative stock.

## Consignment Invoices

Shipments may be invoiced either according to normal consolidation rules (daily, weekly, monthly).

## Consignment Notes

Confirm Consignment will produce a Consignment Note which summarises despatches for the consignment regardless of the stockroom. (Pick notes, if required, will be produced for each stockroom).

The Consignment Note will give brief pack and pallet details where appropriate.

## Pack and Pallet Details

- A 'Pallet' is an outer container; it may be a large crate which packs sit in, or a traditional pallet which packs sit on top of.
- A 'Pack' is the inner container an item is shipped in and the normal 'pack' quantity is used in Pack Rounding.

The system uses some basic rules to produce pack and pallet details for the goods on a consignment. The system creates details for each combination of:

- Item
- Consignment
- Contract
- Stockroom

The system assumes different lot numbers can be mixed on a pallet, but not in a pack, and that different product's cannot be mixed on a pallet, or in a pack. You can later manipulate the pack and pallet details.

Specifications can be made for pack quantities, number of packs per pallet layer, a minimum number of layers and a maximum number of layers, whether complete pallet layers must be shipped, and whether odd packs can be shipped.

The system can generate 'dummy' packs (empty boxes) to fulfil the 'complete layers' flag. There may even be complete 'dummy' layers up to the minimum number of layers.

Pack, Pallet and item weights and volumes are accumulated to calculate consignment totals, which can be maintained prior to confirmation of despatch.

## ASN and ANSI X12 856 Maintenance and Send

ASNs are electronically transmitted messages (usually via EDI) which provide advance information about the configuration of Shipments or consignments. Normally an ASN will contain hierarchical information relating to the shipment itself; the orders or schedules on the shipment; the items contained in the shipment; and how those items are packaged.

Typically the ASN is transmitted to the delivery customer but in certain circumstances may be transmitted to other parties with an interest in the shipment, such as the carrier.

Original Equipment Manufacturers (OEMs) are increasingly placing a requirement upon their suppliers to transmit ASNs to them, often within a specified time span of the shipment actually leaving the suppliers plant. This time span is as small as fifteen minutes for some OEMs.

Many manufacturers wish to use the information contained in the ASN to streamline their Goods Receiving processes. The ASN information may include details of bar-coded packaging labels. Upon receipt, the bar-coded labels are read via hand held scanners and the goods contained within are automatically checked in to stock against expected receipts information provided by the ASN.

The screens relating to item details and their associated packaging details are optional. You can choose how deep into the maintenance options you wish to go by either accepting the packaging information already generated (or not if pack details are not required), or delving deeper and maintaining the packaging details manually.

The ANSI X12 856 ASNs is of increasing importance if you trade in, or with clients in, North America. It behaves in a similar way to a European ASN but differs in its use of the hierarchical levels and does not have a fixed structure. It therefore allows more specific data to be included. Hierarchical levels can be set up for each shipment in any sequence and levels can be omitted. The sequence is agreed between the customer and the supplier.



## Record Locking Considerations

- Company and calendar maintenance require exclusive use of the system.
- The following simultaneous updates are prohibited via record locking:
  - Maintaining Contract Shipping details whilst processing it in Consignment Build.
  - Maintaining a Customer whilst processing one of their contracts in Shipping Processing.
  - Maintaining Despatch Profile / Pack /Pallet / Item whilst processing with them in Shipping Processing.



*Note: The Style Advanced Shipping Notices Company, Customer, and Contract maintenance options will automatically be linked to the corresponding Advanced Customer Scheduling options if this is available, to prevent users having to enter details from two separate menu options when initially creating new Customer and Contract processing defaults.*

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## Chapter 6

### Style EDI Application Interface

### Overview



# Introduction

Electronic Data Interchange is not a new concept. In the early 1960s many large multi-nationals, such as Philips and General Motors, started to transfer internal information and instructions electronically between different application systems, whether by tape or telecommunications.

Since then, technology and business practice have evolved to the point where EDI is virtually a condition of trade in some business sectors.

The application of EDI has accelerated as companies have come to realise that EDI can bring many measurable benefits:

## **Speed**

EDI sends and receives documents quickly. There is none of the delay associated with mailing the documents.

## **Accuracy**

Since EDI transmits documents electronically, there is no need to re-enter the information, thus reducing keying errors.

## **Cost Savings**

The quick turn-around of documents reduces inventory overhead and increases the available cash flow. EDI also reduces mailing costs.

## **Productivity**

Employees are free to perform different tasks. They are no longer needed to enter documents or address, stamp and mail envelopes.

The EDI solution has two closely integrated components. These are:

- The EDI Application Interface module
- Premenos EDI product

Together, these products satisfy the requirements of a successful EDI installation, namely:

- Controlled Application Interfaces
- Standard Message Format Support
- Conversion Facilities
- Communications Management Facilities

EDI will provide support for the most widely-used Message Standards and Value Added Networks. It will also give mailbox access and document conversion facilities. The Application Interface module supplies the facility to import and export documents - in a controlled fashion - to and from the standard Sales Order Processing and Purchase Management modules.

# Overview

EDI Application Interface enables the import and export of documents to and from Sales Order Processing and Purchase Management. These documents, traditionally sent and received in a paper form, can now be exchanged as electronic messages. The application will support:

- Inbound Sales Orders And Order Amendments.
- Outbound Acknowledgements.
- Outbound Advice Of Despatch.
- Outbound Sales Invoices And Credits.
- Outbound Purchase Orders And Amendments.
- Outbound Vendor Schedules.
- Outbound Catalogues.
- Inbound Acknowledgements.
- Inbound Advice Of Despatch.
- Inbound Supplier Invoices (Onto The Accounts Payable Log).
- Inbound Schedule Orders.
- Inbound Advanced Shipment Notices.
- Receive/send order status.
- Receive product activity.

The application can be used in two ways.

The application is designed to cater for the conversion of messages into and out of standard message formats. The Application Interface module adds new function for users of Sales Order Processing and Purchase Management in the shape of direct calls to EDI. So, for example, one interface option will allow the preparation of electronic purchase orders, and will also invoke a routine in EDI to convert those orders into standard message formats for onward transmission to suppliers. This close-coupled link to EDI removes the need for repeated manual intervention in the conversion and transmission process, which can be time consuming, and thus costly.

The second potential use for the interface module is as an electronic batch transaction facility. For example, it is common in certain industries for sales orders to be captured by the field sales force on portable microcomputers. Traditionally, these have then been re-entered onto the central Sales Order Processing application. With the arrival of the Application Interface module, this is no longer necessary. Using existing facilities, the orders can be loaded up onto the host system, and taken into the order bank, in a controlled fashion, using the Application Interface module. There are many similar potential uses for the module.

The control aspect must be underlined. This module applies the same validation logic to inbound and outbound messages as is used in the conventional operation of the system. Full exception and audit reporting is provided.

# Style EDI

EDI provides the following functions:

- Reformatting of messages between the private format of the Application Interface module and the formats of the major international EDI Standards
- Communication modules to enable direct and straightforward connection to all the principal Value Added Networks
- Support for point-to-point communications where a Value Added Network is not in use
- Full transaction logging, error recovery and security features

## **Support Of EDI Standards**

EDI will support the following national and international EDI Standards:

- ANSI X12
- INCOME
- EDIFACT
- ODETTE
- Tradacoms
- TDB
- US

In addition, EDI contains a Standard Modification Tool, which will allow modification of Standards already defined to the product, and definition of new Standards.

## **Value Added Network Support**

EDI is supplied with a module enabling connection to the Premenos Information Network. In addition, further communication modules will be available to enable connection to any, or all of the following Value Added Networks:

- INS/TRIDENT
- AT&T Easylink
- GEIS network services
- McDonnell Douglas network services
- Control Data/Redinet
- Sterling/Ordernet Version 3
- Kleinschmidt network services
- TranSettlement/Transend
- General Motors/GMNET
- Ford/SOLMIS

As well as the Value Added Networks (VAN) listed above, EDI will allow direct point-to-point communications between systems, where a VAN is not used.

## Relationship To Other Applications

The following applications are pre-requisite to the Style EDI Application Interface:

- Inventory Management
- Sales Order Processing (for Sales documents only)
- Purchase Management (for Purchasing documents only)
- Vendor Scheduling (for Vendor Schedules documents only)
- Accounts Payable (for Supplier Invoices)
- Advanced Receiving (for Advanced Shipment Note documents only)
- Application Manager

The application is designed to be able to add translated documents into the Mailbox Send Queue in the EDI application automatically. This requires, however, the trading partner being defined to EDI.

# Application Configuration

To set up the additional data required for sending and/or receiving documents via the EDI Application Interface. The main elements to be created are:

- Documents
- Trading Partners
- Location References
- Translation Maps
- Code Translation Tables

## **Documents**

All document types to be sent via EDI must be defined in the Application Interface, e.g. Purchase Order, Sales Invoice. Within this, the document ID must be specified for each EDI message standard used.

These definitions will be shared by all companies defined to the application.

## **Trading Partners**

All trading partners (customers or suppliers) who are to be communicated with via EDI must be defined in the Application Interface.

Within this, the document types to be sent to each partner must be selected from those defined above. At this point it should be specified if the document is to be printed and/or sent via EDI for this trading partner.

Further details must be defined for the Trading Partner in the EDI application. These are required to support the document translation and communication functions in EDI.

## **Location References**

Where delivery is required to multiple locations, a set of agreed location reference codes must be maintained in order to establish the correct delivery address.

## **Translation Maps**

A translation map is required for each trading partner/document combination, whether they are sent or received. This is used by EDI to map data sent via EDI from the Application Interface database into the appropriate EDI message standard, and vice-versa for data received.

## **Code Translation Tables**

A code translation table is required to convert codes, e.g., Units of Measure, Sales Tax, into the equivalent codes recognised by the appropriate message standard when sending documents.

Conversely, they are used to translate EDI standard codes into codes for receiving documents.

For more details on setting up trading partners see Trading Partners Maintenance.

# Processing Cycle

The following example is provided to illustrate how the EDI processing cycle is supported. The example given is that of an incoming sales order. The procedure for outgoing messages is similar.

Although each step can be executed separately, the products can be configured to allow steps 1, 2 and 3 to be combined into one option leaving step 4 to be executed separately.

## **Step 1 Access Value Added Network**

Having agreed with the applicable electronic trading partners which VAN is to be used to exchange messages, you can define it to EDI. This definition tells EDI how the connection to the VAN should be established.

EDI supports binary synchronous communications with all the networks for which communications modules are provided. For Premenos' IN, full SNA communications are supported. The selected protocols can be used over leased or dialled lines. In the case of dialled connections, either autodial or manual dial is supported. During set-up, other parameters concerning communications with the network will be recorded. EDI stores these parameters with the telephone numbers for dialled connections.

For examples in this product information it is assumed that EDI has established the connections between the AS/400 and an IN.

## **Step 2 Transfer Documents**

Once the communications session is started, EDI will use the previously defined parameters to control the conversation with the network. EDI will then collect any messages sitting in the network mailbox and transfer them down to its own mailbox on the AS/400. It will also check to see if there is any outbound mail (e.g. acknowledgements) that should be sent via IN, and if there are it will transfer them up to the network mailbox. IN will then distribute these to the appropriate trading partners. EDI maintains an on-line log containing the details of each communications session.

Once the transfer process is complete, EDI will - if appropriate - disconnect from the network.

The received message will now be sitting in the EDI mailbox. It can be displayed, printed or deleted while it is there. It can also be translated and moved into the Application Interface Module.

## **Step 3 Translate To In house Format And Files**

Before EDI can successfully translate an incoming order out of its EDI format and into the in-house format, a number of definitions must have been set up within EDI:

- **Trading Partner Transaction Type**

This defines a unique business relationship regarding the two trading partners concerned in the transaction of a particular document, in this case an order. EDI uses this information when handling received messages. The information is also used when addressing and sending outbound messages.

EDI also uses the transaction type to identify the translation map that should be used to translate the document (whether inbound or outbound). The transaction type records other information regarding the relationship that has been agreed with a trading partner, such as the network over which the document should be exchanged, the trading partner's network ID, and version and release values for the standard used.

- **Application Interface File Definitions**

The Application Interface files that will receive the incoming order must be defined to EDI as 'user files'. During translation, EDI directs the translated data to these files.

- **Translation Map**

The relationship between the incoming Tradacoms order and the Application Interfaces files has to be defined. To do this, a 'map' must be created. This map lies at the heart of the translation process. EDI moves data contained in the Tradacoms standard order into the corresponding Application Interface files. A map is required for all EDI documents that are to be sent or received. This is true of all EDI standards.

- **Code Translation Tables**

EDI standards will often have pre-defined codes for certain data elements. These must be translated so as to be accepted as valid. Similarly, if our trading partner uses different values (e.g. UOM, delivery address or product codes), they must also be translated into values acceptable to the particular application. Hence code translation tables. EDI uses these tables to translate standard and trading partner-specific codes into values that are acceptable.

The translation process takes all these elements into account. When completed, the translated order will be written to the appropriate Application Interface file.

#### **Step 4 Application Interface**

Once EDI has successfully translated the incoming order and placed it in the Application Interface file; an interface option can be executed to place it on the Sales Order Processing order bank. This interface program will perform the same validation upon the incoming orders that is applied normally to orders entered to the system via the keyboard. Most exceptions will cause the order to be suspended for further investigation. In the case of major exceptions, the order will be rejected. Full exception and control reports are provided.

The logic described in this example is applied to all the documents (whether inbound or outbound) supported by EDI and the EDI Application Interface. In the case of outbound documents (e.g. purchase orders), most of the validation will already have been carried out by the application generating the document. For outgoing documents, EDI and the Application Interface examine the validity of the document as an EDI message, deciding whether they should be sent via EDI and/or printed.



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## Chapter 7

### Style Production Overview



# Introduction to Style Production

Style Production has been developed to meet specifically the production needs of organisations within the clothing and apparel, footwear and soft furnishings industries.

Although the products that these organisations produce are diverse, the manufacturing requirements are similar. They require:

- A horizontal sequence of operations defined in the route:



- A means of maintaining information about products at a generic level.
- Two levels of items, although it is possible to have more than two levels:
  - Styles. The finished goods.
  - Materials. The materials that go into producing the styles.

Style Production provides a means of maintaining common information for the multiple variations of a style in one action. That is, you do not have to enter information about each style variant separately.

In addition, Style Production offers the option to produce some style variants on different machines and/or using different materials.

In common with other JBA applications, you can operate a single copy of Style Production for any number of companies.

Using control parameters held on the Company Profile and the Organisational Model, you can configure the characteristics of each company.

# Brief Overview of Style Production

Style Production comprises four integrated applications, as shown below:

| Application Code | Application                            | Application Product Guide              |
|------------------|----------------------------------------|----------------------------------------|
| P1               | Style Production Definition Management | Style Production Definition Management |
| P2               | Master Production Scheduling           | Style Production Planning              |
| P3               | Material Requirements Planning         | Style Production Planning              |
| P4               | Style Production Control               | Style Production Control               |

The Company Profile governs these core applications.

## **Style Production Definition Management (PDM)**

Style Production Definition Management (PDM) offers a self-contained environment for defining and managing products, processes, resources and costs.

Use the maintenance functions of PDM to set up information that is critical to the way in which Style Production functions. Once you have set up this data, PDM can generate product costs (both standard and non-standard) and provide an extended analysis of product definitions and processes.

You can also choose to run PDM stand-alone.

## **Forecasting**

Forecasting belongs to the PDM application.

Forecasting allows you to forecast demand down to the level of individual variants. For example, for clothing you can forecast down to the level of style/colour/size.

You have the option to group products into product families.

You can input the forecast demand into MPS and MRP processing.

## **Master Production Scheduling**

Master Production Scheduling (MPS) provides a top level master production schedule for manufacturing styles.

## **Material Requirements Planning**

Material Requirements Planning (MRP) provides a detailed material plan for your business.

### **Production Control**

Production Control controls and maintains all aspects of the production cycle.

The cycle begins with the release of suggested production orders from MPS. Alternatively, you can create production orders yourself. The cycle continues through material issuing and work-in-progress bookings. At the end, Inventory receives the finished goods.

Production Control also provides comprehensive enquiries and reports to monitor the progress of individual production orders and the performance of factory/production units.

# Interfaces

There is one mandatory interface to Style Production and that is Style Inventory Management.

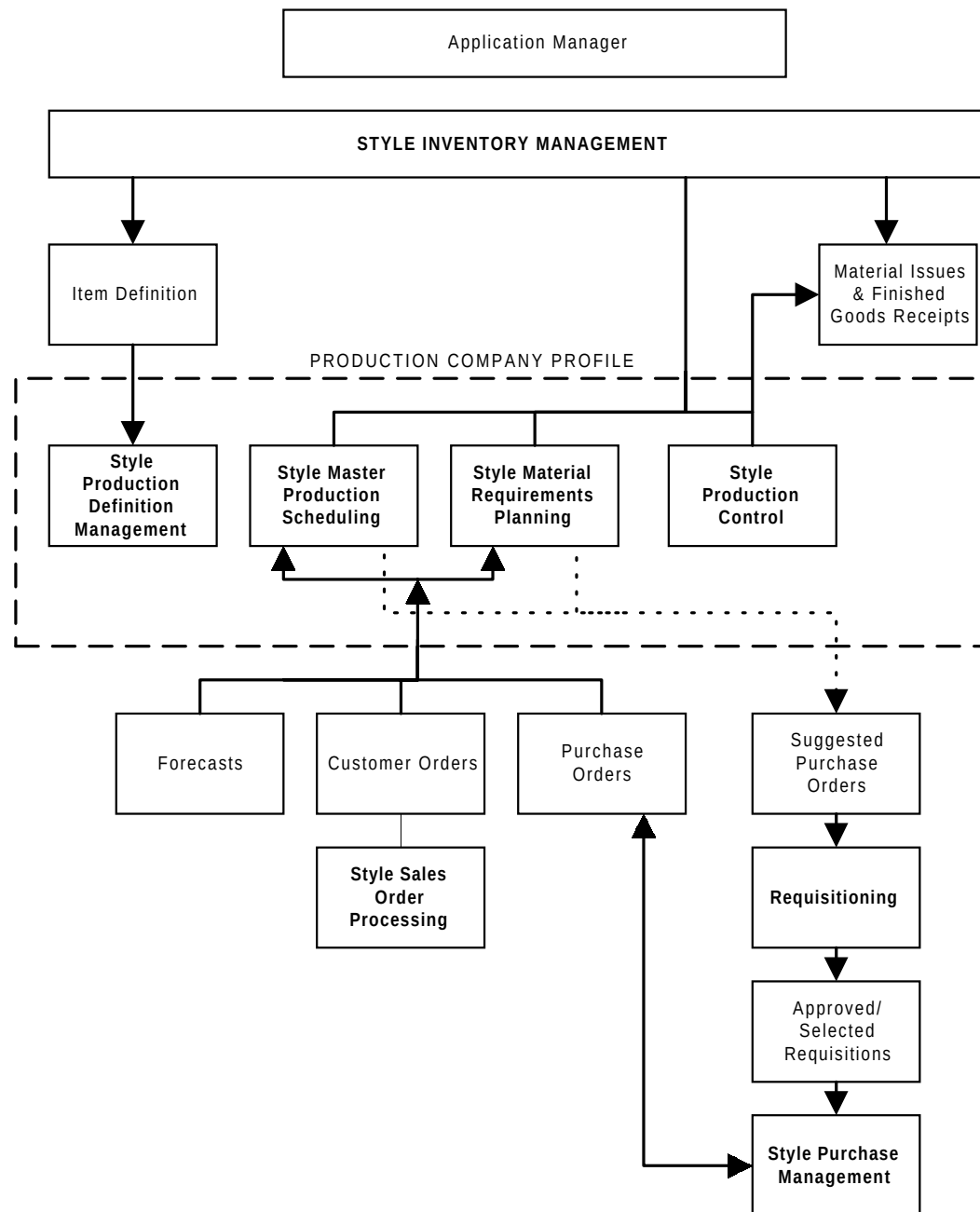
The following interfaced applications are optional:

- Style Purchase Management. However, if you intend define subcontracted operations on route/BOMs, then Style Purchase Management is mandatory.
- Style Sales Order Processing.
- Warehousing.

To use certain facilities available within the system, additional optional applications must be 'live'. This is especially true within Style Production Planning which looks to other areas of a fully integrated production system for external demand and supply information.



For a full description of each interface, please refer to the application's own product guide.



*Figure: The Style Production System*

## Optional Interfaces

Style Production can function fully when interfaced to Style Inventory Management alone. More commonly, Style Production operates within a larger, integrated systems environment.

Integration with the following applications provides additional resources for management of a sophisticated, closed loop production environment.

### **Style Purchase Management**

Style Purchase Management and its associated Requisitions function provides a two way interface to Style Production:

- Style Purchase Management provides MPS/MRP with details of current material supply information.
- MPS/MRP generates suggested purchase orders, which when confirmed, raise purchase requisitions, which in turn are converted to firmed purchase orders.

You can arrange to create purchase orders automatically for subcontracted operations.

Complementary to running Style Purchase Management is the interface to Accounts Payable (Purchase Ledger), where you must create all supplier records.

Subcontract operations depend on Purchase Management, as each operation designated subcontracted must be assigned a supplier, which in turn must be valid within Purchase Management and associated with a live account in Accounts Payable.

### **Style Sales Order Processing**

Style Sales Order Processing provides a two way interface to Style Production:

- Style Sales Order Processing provides MPS/MRP with details of current sales demand information.
- Style Production provides Sales Order Processing, via Style Inventory Management, with current details of how supply is intended to meet sales demand.

Style Production depends on Style Sales Order Processing for the supply of demand information only.

To create a forecast of sales demand, use Style Production. Alternatively, use the System 21 Forecasting application to extract information about styles from Style Inventory and sales history from Style Sales Analysis.

### **Warehousing**

Warehousing is an extension of Style Inventory Management stockrooms. You can use Warehousing to designate location details for materials and styles within a stockroom.

You can use Warehousing to:

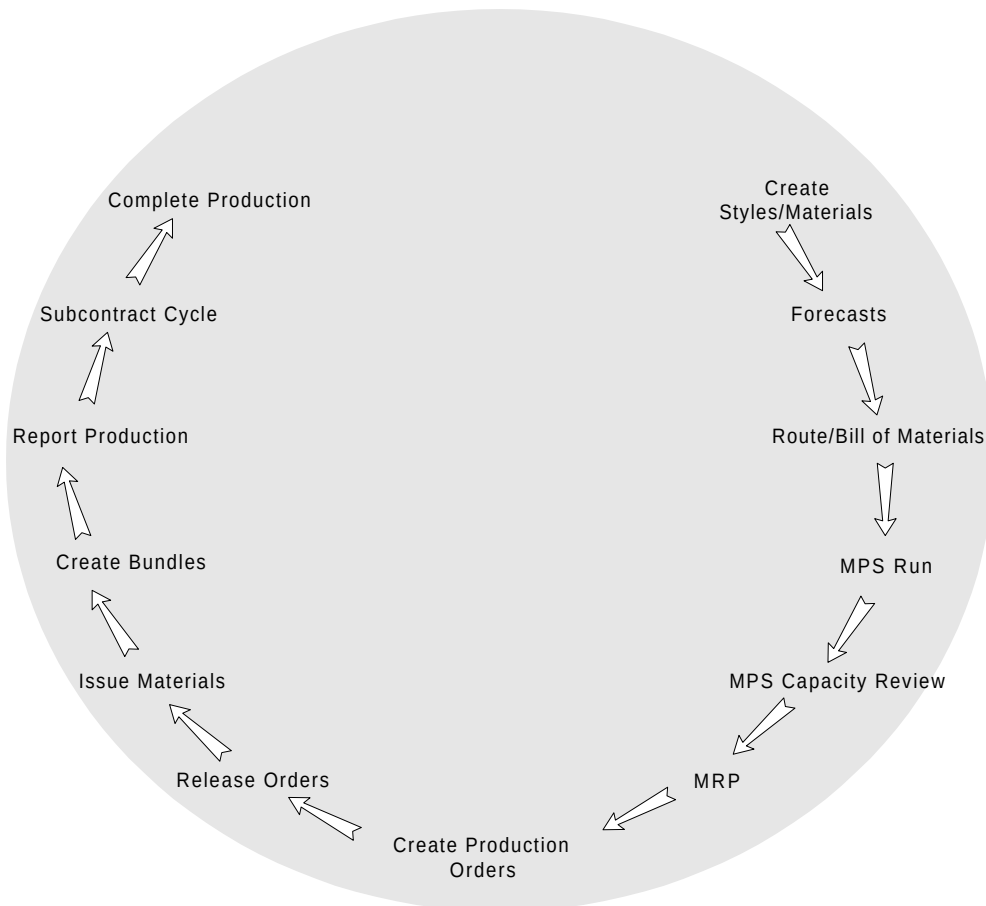
- Satisfy material requirements generated by production orders.
- Provide a receiving location for finished goods.

Warehousing is not a pre-requisite of Style Production.

# The Business Cycle

## Introduction

The following diagram gives an overview of the business cycle:



## Create Styles And Materials

Styles are the end products that your business sells to customers. Materials are the materials, such as fabric, trims and packaging, that make up the styles.

With Style Production, you can enter information about a style or material in one go. That is, you do not have to enter information for each colour and size variant separately.

Style views styles and materials as multi-dimensional. The first dimension is always the style or material itself. You can use up to four dimensions per style, as follows:

- Style.
- or
- Style / Colour.
- or
- Style / Colour / Size.

or

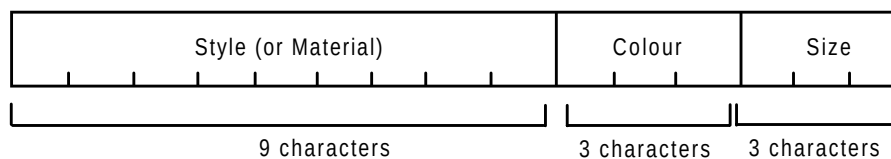
- Style / Colour / Size / Fit.

| Dimension           | Characteristic | Such as             |
|---------------------|----------------|---------------------|
| Style (or Material) | -              | Skirt               |
| 2nd Dimension       | Colour         | Navy, Black, Green  |
| 3rd Dimension       | Size           | 10, 12, 14          |
| 4th Dimension       | Fit            | Short, Medium, Long |

Therefore, each style and material can have colour variants and up to two types of size variant. Size variants can take many forms such as:

- Trousers: inside leg and waist size.
- Curtains: height and drop.
- Shoes: size and fit.

Each style and material has a unique identifier, known as the style code or material code. Style variants extend this identifier to include codes for the colour and size(s). A variant code has the following format:



Examples of typical codes for style and material variants are shown below:

|                       | Dimensions            | Variant                                                                | Typical style/material code |
|-----------------------|-----------------------|------------------------------------------------------------------------|-----------------------------|
| <b>1 dimensional:</b> | Style                 | There are no variants: the label has no colour or size characteristics | LABEL005                    |
| <b>2 dimensional:</b> | Style/Colour          | Green scarf                                                            | SCARF01 GN                  |
| <b>3 dimensional:</b> | Style/Colour/Size     | Large black tee shirt                                                  | TS15G BLK L                 |
|                       |                       | Red fabric 140cm                                                       | FAB001 RD 140               |
| <b>4 dimensional:</b> | Style/Colour/Size/Fit | Navy skirt size 12 short                                               | SK010 NV 12S                |

## **Forecasts**

You can enter forecasts at the following levels:

- Style.
- Style / Colour.
- Style / Colour / Size.
- Style / Colour / Size / Fit.
- Product Family.

After entering a forecast, you can spread the forecast as shown below:

| FROM            | TO                    |
|-----------------|-----------------------|
| Product Family: | Style                 |
|                 | Style / Colour        |
|                 | Style / Colour / Size |
| Style:          | Style / Colour        |
|                 | Style / Colour / Size |

## **Routes / Bills Of Material**

A production route can produce multiple variations of styles such as different colours and sizes of jumpers, shirts, pairs of shoes. A production route can also produce a component for use in the manufacture of a finished style.

A production route defines:

- The operations required to make a style.
- The materials necessary to complete the operation.
- The machines at which the work is carried out.

Each style can have one or more routes. On each route, you can:

- Match style colours to material colours.
- Change material quantities per style size.
- Change operation details for different style colours and sizes.

Style Production caters for multiple plants. If you are making a particular style in more than one plant, multi-plant planning can tell Style Production to shift the load from one factory to another factory. This facility is useful where a factory already has a full load.

You can experiment with model planning routes that do not affect live production. You can create model planning routes from new, or you can copy a live production route to a model planning route. When you are happy with the changes you have made to a model planning route, you can copy the route back to replace a live production route.

You can control access to live production route maintenance. For example, you can authorise a user to maintain model planning routes with no access to live production routes.

### **MPS Run**

You can run MPS centrally or multi-plant or from a single plant.

MPS always performs a full regenerative run.

An MPS run carries out the following activities:

1. Examines sales orders, purchase orders, production orders, stocks and forecasts.
2. Nets off sales orders against forecasts to produce a net demand.
3. Works out when styles are required in terms of time buckets.
4. Suggests new/changed production orders and purchase orders. New orders can be based on sourcing rules and planning overrides.
5. Through capacity planning, determines whether there is sufficient capacity to action the suggested production orders.
6. Generates machine and labour loading.



Tip: You can arrange for MPS to consider purchased styles also, without running MRP. MPS can suggest purchased orders which you then need to activate via Requisitioning in Style Purchase Management.

### **MPS Capacity Review**

After an MPS run, you can review MPS and you can review capacity loading.

During the MPS review, you can:

- Review production planning at style level.
- From a suggested production order, drill down to individual style variants.
- Change the route, date and quantity on a suggested production order.
- Change the status of a suggested production order to planned or confirmed.
- Review the impact of your changes on capacity.

In the garment industry, the most critical planning element is labour skills. Therefore, Style Production differs from Standard Production by including skill-based planning.

During a capacity review, you can:

- Review the labour and machine loads that MPS generates.
- If you see an overload, move a suggested production order before you firm up the order.
- Consider how the move affects labour and machine load.
- Manipulate the data again.

- When you are happy, firm up the order and save the model as a different model.

### **MRP**

You can run MRP either before or after firming up MPS orders.

You can perform a full regenerative run or a selective run. You can also MRP to consider only the changes that have occurred since the last MRP run.

Therefore:

1. Run MPS.
2. Stop and manipulate the data.
3. Run MRP.



*Note: MRP does not carry out capacity planning.*

### **Create Production Orders**

Create production orders manually or process production orders suggested by MPS.

MPS suggests production orders for styles broken down to style variant level. MPS calculates supply at style variant level, but presents information at style summary level so that you can manipulate the information at style level.

### **Release Orders**

To make sure you have all the materials you require, you can carry out a trial kit. Within Style Production, you can make lot control automatic. You can also arrange to create bundles at order release.

If a subcontractor is responsible for any of the operations, then Style issues a purchase order.

### **Issue Materials**

When you issue materials, you can:

- Accept or change the lots that Style allocates.
- Issue substitute materials.
- Record unplanned issues.
- Transfer materials to a subcontractor, if you are outworking the cutting operation.

### **Create Bundles**

A bundle is a large plastic bag containing all the fabrics and other materials (eg buttons) that go into producing a garment.

You can create bundles in two ways:

- Automatic. You can set up a number of different bundle types for which you specify the quantity to go in each bundle. You can allocate a bundle type to a specific route and arrange for Style to generate bundles automatically.
- Manual. You can decide on the bundle size when you create the bundle.

The bundling process can print barcoded bundle tickets. Style Production uses bundle tracking to track the garment from one operation to another.

### **Report Production**

There are various ways of entering time spent in production:

- Operator bookings. Actual time is keyed in.
- Timesheet bookings. Actual time is calculated.
- Scan in barcodes on bundle tickets. Alternatively, key in the bundle number. Actual time is calculated.

### **Subcontract Reporting Cycle**

Where several subcontractors are carrying out consecutive operations, you can define the operations separately. Style can report on progress from operation to operation.

When you release the production order, Style looks down the list of subcontracted operations and generates a subcontractor purchase order.

When you report the operation previous to the subcontracted operation, Style creates a subcontract shipper against which you can ship out the goods. If several subcontractors are on the route, you can issue either one or several subcontract shippers.

When you receive the goods back, Style generates a good received note for the purchase order. You can then carry on booking against the next operation on the route.

### **Complete Production**

You can complete production when you book in the final operation. If you book in the full quantity, Style automatically completes the production order.

Even if the quantity you book in is short, you can close the order.

You can re-open a production order.

At the end of the production cycle, the completed products pass into Style Inventory.

## The Critical Path to Production

This section provides an overview of the critical path to follow in setting up a Style Production company, managing data and ultimately shipping your finished styles to a customer.

The following path is relevant to a fully comprehensive integrated system that is running all of the Style Production applications and also the following applications:

- Style Inventory Management.
- Style Sales Order Processing.
- Style Purchase Management and, by implication, Accounts Payable and Purchase Ledger.

If your system is not as complex as this, then you may skip non-relevant stages and data definitions.

Each of the following stages is numbered consecutively in the order in which data definition should take place.



Tip: Use the critical path listed below as a guide only. You may be able to jump to another task before all previous tasks are complete.



Tip: You may want to hold discussions with your system consultant with regard to an implementation procedure specific to your requirements.



Tip: For a full description of each data element in the critical path, refer to the associated product guides that accompany each System 21 application.

| The Critical Path to Production |                                                                                                          |                  |                                        |
|---------------------------------|----------------------------------------------------------------------------------------------------------|------------------|----------------------------------------|
|                                 | Description                                                                                              | Applica-<br>tion | Menu Option                            |
|                                 | <b><u>Prior To Style Production</u></b>                                                                  |                  |                                        |
| 1                               | Create a new company(s) via the Applications Manager                                                     | Admin Functions  |                                        |
| 2                               | Authorise user(s) to new company in Style Distribution                                                   | Admin Functions  |                                        |
| 3                               | Copy company in Style Distribution                                                                       | STIN             | Copy Company                           |
| 4                               | Maintain Style Sales Order Processing, Company Profiles. Set the Style Production in Use flag to 1 (yes) | STIN             | Sales Order Processing Company Profile |

|    |                                                                                                                                                                   |                       |                                      |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------|
| 5  | Maintain Style Inventory Management, Company Profiles. Create stockroom records, including work-in-progress (WIP) locations and subcontractor materials stockroom | STIN                  | Inventory Management company profile |
| 6  | Maintain Style Purchase Management Company Profiles                                                                                                               | STIN                  | Purchase Management Company Profile  |
| 7  | Create suppliers, including subcontractors                                                                                                                        | PL (Accounts Payable) | Names and Addresses                  |
| 8  | Create colours for each company                                                                                                                                   | STIN                  | Colours                              |
| 9  | Create Inventory Management calendars                                                                                                                             | STIN                  | Calendars                            |
| 10 | Create Inventory Management periods                                                                                                                               | STIN                  | Period/Week End Control              |
| 11 | Create search families                                                                                                                                            | STIN                  | Search Family Codes                  |
| 12 | Create sizes                                                                                                                                                      | STIN                  | Sizes                                |
| 13 | Create size masks                                                                                                                                                 | STIN                  | Size Masks                           |
| 14 | Create inventory descriptions                                                                                                                                     | STIN                  | Descriptions                         |
|    | <b>Style Production</b>                                                                                                                                           |                       |                                      |
| 15 | Authorise user(s) to Style Production                                                                                                                             | Admin Functions       |                                      |
| 16 | Maintain Style Production Company Profiles                                                                                                                        | P1                    | Maintain Company Profile             |
| 17 | Create Organisational Models                                                                                                                                      | P1                    | Organisational Models                |
| 18 | Create Company parameters                                                                                                                                         | P1                    | Maintain Parameter File              |
| 19 | Create styles and materials                                                                                                                                       | P1                    | Style/Material Details               |
| 20 | Create cost centres                                                                                                                                               | P1                    | Cost Centres                         |
| 21 | Create calendars                                                                                                                                                  | P1                    | Calendars                            |
| 22 | Create shift profiles                                                                                                                                             | P1                    | Shift Profiles                       |
| 23 | Create departments                                                                                                                                                | P1                    | Departments                          |
| 24 | Create machines                                                                                                                                                   | P1                    | Machines                             |
| 25 | Create work centres                                                                                                                                               | P1                    | Work Centres                         |
| 26 | Create labour skills                                                                                                                                              | P1                    | Labour Skills                        |
| 27 | Create labour profiles                                                                                                                                            | P1                    | Labour Profiles                      |
| 28 | Create bundle ticket types                                                                                                                                        | P1                    | Bundle Ticket Types                  |
| 29 | Create resource capacity profiles                                                                                                                                 | P1                    | Resource Capacity Profiles           |

|    |                                                     |                       |                                  |
|----|-----------------------------------------------------|-----------------------|----------------------------------|
| 30 | Create critical resources                           | P1                    | Critical Resources               |
| 31 | Create standard operations                          | P1                    | Standard Operations              |
| 32 | Create style route/BOMs                             | P1                    | Style Route/Bill Of Material     |
| 33 | Create price list profiles                          | STOE                  | Price/discount profiles          |
| 34 | Create customers                                    | STOE                  | Customers                        |
| 35 | Create price lists                                  | STOE                  | Price Lists                      |
| 36 | Create discount lists                               | STOE                  | Discount Lists                   |
| 37 | Create depot profiles                               | STOE                  | Depot Profiles                   |
| 38 | Enter sales orders                                  | STOE                  | Order Entry                      |
| 39 | Create MPS planning models, part one                | P2                    | Maintain Model Stockrooms        |
| 40 | Create MPS planning models, part two                | P2                    | Maintain Reporting Profile       |
| 41 | Create style forecasts                              | P1                    | Maintain Style Forecast          |
| 42 | Spread forecasts to variants                        | P1                    | Generate Variant Forecast        |
| 43 | Run MPS                                             | P2                    | Run Production Scheduling        |
| 44 | Review MPS                                          | P2                    | Review Production Schedule       |
| 45 | Review capacity                                     | P2                    | Capacity Review                  |
| 46 | Confirm suggested orders                            | P2                    | Confirm Suggested Orders         |
| 47 | Create MRP planning models, part one                | P3                    | Maintain Model Stockrooms        |
| 48 | Create MRP planning models, part two                | P3                    | Maintain Reporting Profile       |
| 49 | Run MRP                                             | P3                    | Run Material Planning            |
| 50 | Review MRP                                          | P3                    | Review                           |
| 51 | Create purchase orders from suggestions, part one   | STPM (Requisitioning) | Create Requisitions from MPS/MRP |
| 52 | Create purchase orders from suggestions, part two   | STPM (Requisitioning) | Requisitions Confirmation        |
| 53 | Create purchase orders from suggestions, part three | STPM (Requisitioning) | Requisitions Authorisation       |
| 54 | Create purchase orders from suggestions, part four  | STPM (Requisitioning) | Create Purchase Orders           |
| 55 | Create purchase orders from suggestions, part five  | STPM                  | New Orders                       |

|    |                                          |                       |                                                                                      |
|----|------------------------------------------|-----------------------|--------------------------------------------------------------------------------------|
| 56 | Enter purchase orders manually, part one | STPM                  | Order Entry                                                                          |
| 57 | Enter purchase orders manually, part two | STPM                  | New Orders                                                                           |
| 58 | Enter raw material receipts              | STPM                  | Goods Receiving                                                                      |
| 59 | Transfer purchasing costs                | P1                    | Purchased Items Re-Cost                                                              |
| 60 | Full cost roll up                        | P1                    | Item Master File Re-Cost                                                             |
| 61 | Create activity types                    | P1                    | Control File Maintenance                                                             |
| 62 | Create operators                         | P1                    | Operators                                                                            |
| 63 | Create teams                             | P1                    | Teams                                                                                |
| 65 | Define user authority levels             | STPM                  | User Authority                                                                       |
| 66 | Create manual production orders          | P4                    | Maintain Orders                                                                      |
| 67 | Allocate lot materials                   | P4                    | Order Progress                                                                       |
| 68 | Release production orders                | P4                    | Release Orders                                                                       |
| 69 | Issue materials                          | P4                    | Issue Materials                                                                      |
| 70 | Report (book) production                 | P4                    | Production Order Booking<br>or<br>Operator Booking<br>or<br>Manual Timesheet Booking |
| 71 | Enter subcontractor WIP shippers         | P4                    | Subcontractor WIP Shipper                                                            |
| 72 | Progress subcontractor work              | P4                    | Progress At Subcontractor                                                            |
| 73 | Receive from subcontractor               | P4                    | Receive from Subcontractor                                                           |
| 74 | Receive into finished goods              | P4                    | Production Order Booking<br>or<br>Operator Booking<br>or<br>Manual Timesheet Booking |
| 75 | Enter supplier invoices, part one        | PL (Accounts Payable) | Item Entry                                                                           |
| 76 | Enter supplier invoices, part two        | PL (Accounts Payable) | Confirm Log Entries                                                                  |
| 77 | Enter supplier invoices, part three      | PL (Accounts Payable) | Volume Log to Ledger Transfer                                                        |

|    |                        |      |                            |
|----|------------------------|------|----------------------------|
| 78 | Match invoices to GRNs | STPM | Match Invoices to Receipts |
| 79 | Despatch to customer   | STOE | Confirmation of Despatch   |

- 
- 
- 
- 
- 

## Chapter 8

### Style Production Definition

### Management Overview



# Introduction

Production Definition Management, shown in the figure below, manages the Style Production database. This database contains:

- Extended production details of all the styles and materials used within Style Production.
- Information about resources (for example, labour skills and machines) that you use to combine materials into finished products.
- Facilities to cost products.

Style Inventory Management maintains information about styles and materials, both purchased and produced. Therefore, you must define in Style Inventory Management every style and material that you wish to use within Style Production.



Note: Before implementing Style Production within your system, make sure you have a clear understanding of the database, its links to Style Inventory Management and the company profile.

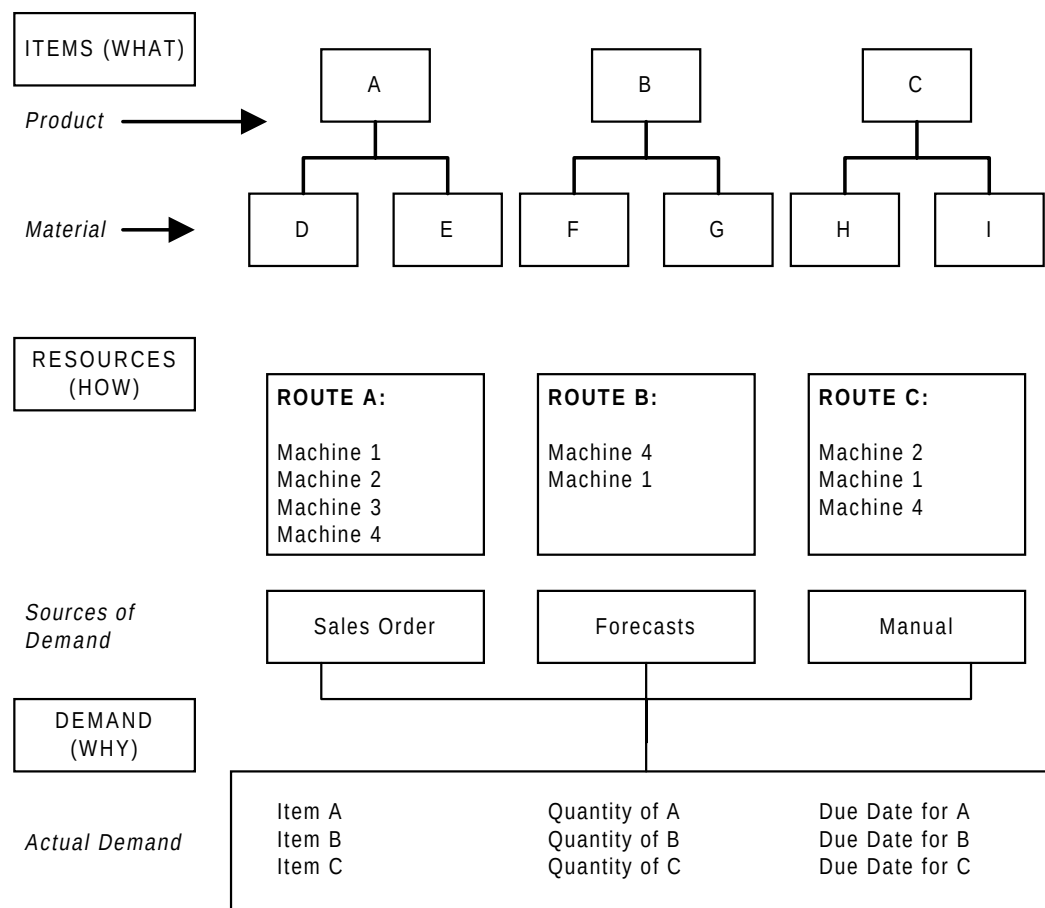


Figure: Production Definition Management

## Resources

Resources are classified as:

- Machines, which you can group into work centres. The assigned cost centre defines the cost basis of each machine.
- Departments.
- Labour skills and profiles.
- Operators and teams.
- Subcontractors.

# Company Profile

Style Production, in common with all JBA Business applications, operates within a multi-company environment.

For each production company, you need to create a company profile to:

- Set the basic system defaults and operation policies relevant to the company.
- Define the costing elements.
- Define base calculation parameters used within Style Production.



*Note: Access to the company profile is usually restricted to system managers and implementation project managers who require a comprehensive understanding of the implications of each parameter setting. However, all users need to be aware of the purpose and features of the Company Profile.*

In addition, to specify how you wish to carry out production at each plant, you can create organisational models for your plants.

## **Cost Elements**

The Style Production company profile provides 14 cost elements for you to use to build up and analyse item unit costs. Ten elements are standard within Style and four elements are user-defined.

# Defining Production Capacity

This section looks at the parameters and resources that describe the production capacity available in your environment.

Within this context, capacity is synonymous with time. That is,

1. How many working days are there available in a week and in a year?
2. How many hours does it takes to make a product?

When you have established these two factors, you can determine whether you have enough, too little or too much (that is, spare) capacity.

## Production Calendars

A production calendar defines how many working days there are in a year. You can define one calendar as the company default to be applied on all occasions or you can define more specific calendars.

If there are variations in the number of days different machines and/or work centres operate within your environment, use multiple calendars. For example:

| Calendar    | Scenario                                                                         | Working Days                                                        | Non Working Days                                                                     |
|-------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Calendar 01 | Default calendar                                                                 | Monday to Friday                                                    | Weekends, bank holidays and national holidays.                                       |
| Calendar 02 | For machines that cannot stop, such as knitting machines                         | Every day                                                           | -                                                                                    |
| Calendar 03 | A particular work centre produces swimwear in anticipation of the summer season. | Monday to Friday<br><br>Weekends during January, February and March | Bank holidays and national holidays.<br><br>Weekends from April through to December. |
| Calendar 04 | Every 6th week a particular machine is shut down for maintenance.                | Weeks 1 to 5, 7 to 11, ...                                          | Weeks 6, 12, ...                                                                     |

# Machine Capacity

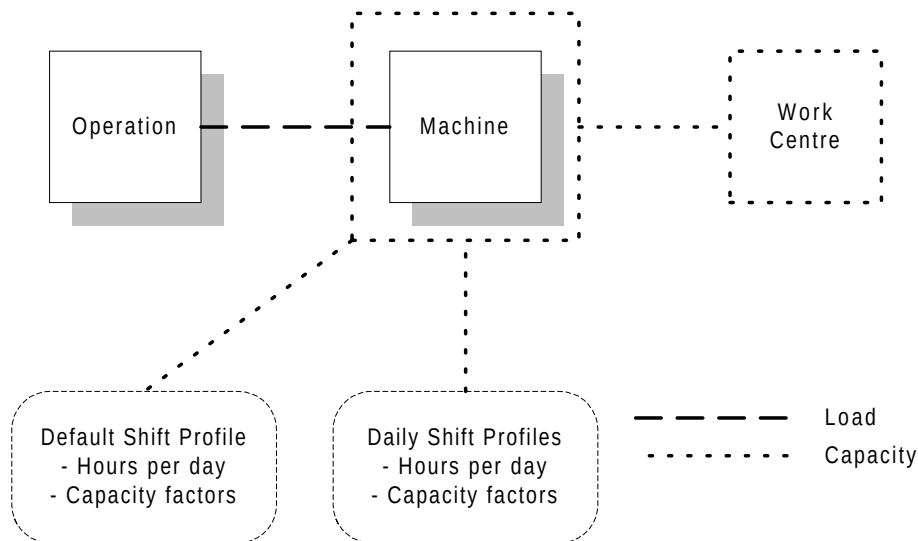


Figure -: Machine Capacity & Load

First of all, decide whether you wish to set up machines as single or multiple resources:

- Single Resource. Define each machine individually.
- Multiple Resource. Group machines of a similar type into a work centre. For example, you may wish to group all your knitting machines as a multiple resource.

You must book each operation on a production route to one machine or one work centre.

To load work across a number of machines without grouping those machines into a work centre:

1. Define a separate route for each machine.
2. Load production orders manually to those routes.

You can either specify the hours for each machine directly or allocate a shift profile.

## **Shift Profile**

A shift profile is simply a list of start and finish times, identified by numbers, for each work shift.

When creating a shift profile, you can enter a new shift pattern after the current pattern to come into force on a specified future date.



*Note: There is no default profile for a working day, this must be explicitly defined against the machine and shift profile codes.*



***Warning: The end date of the calendar is determined by the start date entered and the number of days defined in the Calendar section of the Company Profile. Therefore, set the Company Profile to allow creation of calendars for 52 and 53 weeks.***

# Labour Capacity

To be able to review labour load, you should:

- Define departments.
- Define labour skills.
- Define labour profiles, combining labour skills.
- Assign labour profiles to operations.

Style can then:

- For each operation, determine a set up time and an operation time.
- Determine the length of time each skill level is required at set up and run time and the required number of operators.
- Determine capacity requirements at department and skill level.

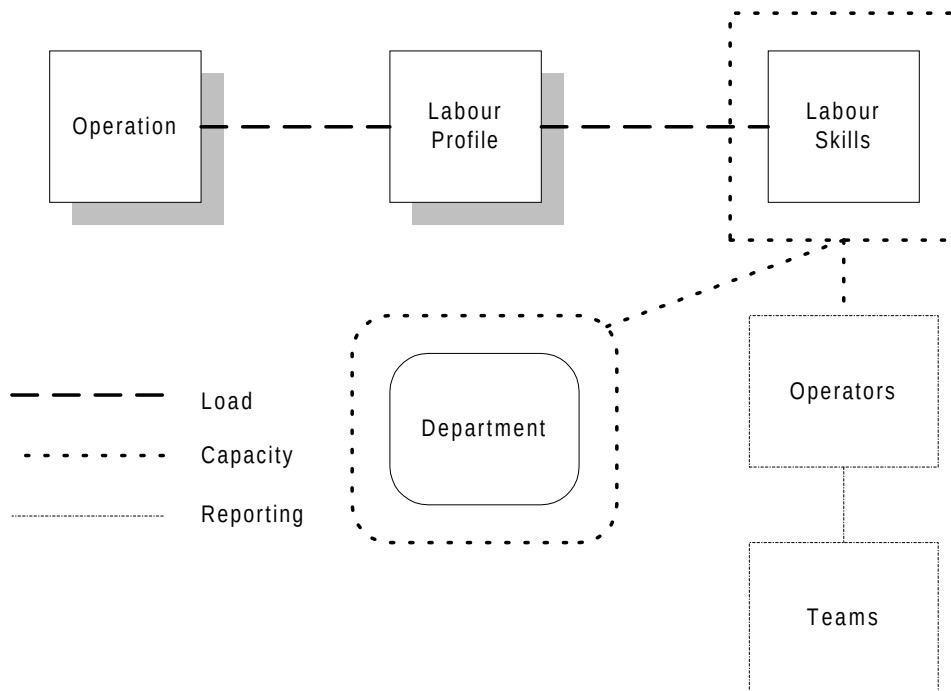


Figure -: Labour Capacity & Load

Labour hours per day are held by department, along with absenteeism, utilisation and average performance percentages.

Load is calculated at labour skill level by using the labour run times and the labour profile attached to an operation.

Capacity is calculated by:

- Adding up the operators attached to a labour skill and department on a given day.

- Using the standard hours per day, absenteeism, utilisation and standard performance values associated with the department in the following calculation:

$[\text{Std Hrs/Day} - (\text{Std Hrs/Day} * \text{Absenteeism})] * \text{Utilisation} * \text{Average Performance}$

Operations can be reported at operator or team level.

## Scheduling Work

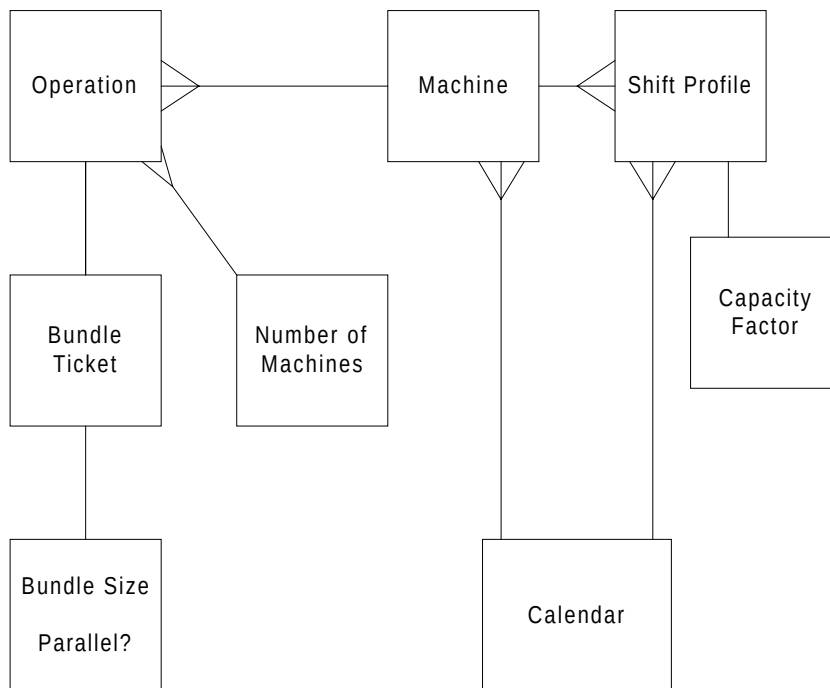


Figure -: Scheduling Work

Work is scheduled using the following procedure:

Firstly, calculate the available shift hours per day:

- Shift profile hours (effectivity date controlled).

OR

- Total of shift lengths.

OR

- Default to 8 hours.

Secondly, calculate the number of machines to be used in parallel; this is the lowest of:

- Number of machines on operations.
- Number of bundles on order:

$$\left( \frac{\text{Order Quantity}}{\text{Default Bundle Quantity}} \right)$$

- Capacity factor if shift profiles exist.

$$\left( \frac{\text{Shift Capacity}}{\text{Shift Length}} \right)$$

Thirdly, if no shift profiles exist:

- Calculate available hours on machine for the order

(Available Shift Hours \* Number of Machines)

# Machines

You can define a machine in either of the following ways:

- As an individual resource.
- To represent a logical view of total resource capacity.

Capacity analysis looks at machines to establish whether they have sufficient capacity to fulfil a proposed production schedule.

Machines may be assigned to:

- Operations. You need to associate either a machine or a work centre with each operation. Then, Style can develop work schedules and identify the resources that are available to carry out the work.
- Work Centres. Optionally, use work centres to group machines for different views of capacity requirements and constraints.



Tip: You can create a summary capacity route for a style, with each operation linked to a machine that represents the total capacity of a group of similar resources.

## **Standard And Maximum Capacity And Efficiency**

You can define a machine's standard capacity and maximum capacity:

- Standard capacity indicates the default number of hours available on a working day. That number may vary between different days in the week.
- Maximum capacity indicates the number of hours that can be made available on a day. For example, an extra shift might be introduced or a shared or redundant resource might become fully available. The maximum capacity is only effective on working days defined on the production calendar. To introduce additional days, such as weekend working, you would need to change the current calendar or apply a new calendar.



*Note: Capacity planning uses standard capacity when scheduling work at a machine. To reschedule using maximum capacity, you must change the standard capacity values and re-run the scheduling process.*

## **Taking Account Of Performance Differences**

Individual machines vary in performance. When comparing scheduled load to available capacity, you can apply a standard efficiency factor. You can use this factor where there are noticeable performance differences between machines, especially in groups or cells.

For example, if you have defined a standard operation time of 60 minutes, then machine 001 may actually take 72 minutes to complete the operation and machine 002 may take only 55 minutes. An efficiency factor can take account of machine 002 completing ahead of machine 001 when processing the same work.

## Calculating Hours Per Day Per Machine

A production calendar associated determines the number of days a machine is operational throughout the year. By default, this is the calendar on the Company Profile.

There are four methods of calculating the number of hours available on working days. These methods are listed below in ascending order of complexity:

- 8 hours default.
- Shift details.
- Default shift profile.
- Shift/capacity profile.

The following tables illustrate, in ascending order of complexity, the four methods of defining working hours for machines over one working week.

- The examples use a default production calendar (01).
- A thick line border encloses the values set for each machine.
- The Std Capacity and Max Capacity columns indicate the resulting machine capacity on a day to day basis.
- The totals boxes at the foot of each table show the standard and maximum capacity of the machine calculated over one working week.

### **8 Hours Default**

If no time is defined for the machine, Style assumes a default of 8 hours for both standard and maximum capacity for every working day within the year.



*Note: You cannot use this method for multiple resources.*

| Production Calendar - 01 |           |  |  | Machine A |              |           |              |
|--------------------------|-----------|--|--|-----------|--------------|-----------|--------------|
| Date                     | Work Day? |  |  | Std Hours | Std Capacity | Max Hours | Max Capacity |
| Jan 1 Mon                | Y         |  |  | 8         | 8.00         | 8         | 8.00         |
| Jan 2 Tue                | Y         |  |  | 8         | 8.00         | 8         | 8.00         |
| Jan 3 Wed                | Y         |  |  | 8         | 8.00         | 8         | 8.00         |
| Jan 4 Thu                | Y         |  |  | 8         | 8.00         | 8         | 8.00         |
| Jan 5 Fri                | Y         |  |  | 8         | 8.00         | 8         | 8.00         |
| Jan 6 Sat                | N         |  |  |           |              |           |              |
| Jan 7 Sun                | N         |  |  |           |              |           |              |
|                          |           |  |  |           | 40.00        |           | 40.00        |

### **Shift Details**

You can define a standard and maximum capacity for up to three shifts per working day. When totalled, these give the number of hours available per working day.

In addition to shift lengths, you can also define a standard and maximum capacity that applies to every working day within the year. This is used to indicate that a machine represents more than one resource.

For example, if Machine A represents 2 sewing machines that operate 8 hours per day, then the standard capacity is 16 hours.



*Note: With this method, you have to change capacity availability individually for each machine..*

| Production Calendar - 01 |           |  |  | Machine A |              |           |              |
|--------------------------|-----------|--|--|-----------|--------------|-----------|--------------|
| Date                     | Work Day? |  |  | Std Hours | Std Capacity | Max Hours | Max Capacity |
|                          |           |  |  | 8         | 16.00        | 16        | 32.00        |
| Jan 1 Mon                | Y         |  |  | 8         | 16.00        | 16        | 32.00        |
| Jan 2 Tue                | Y         |  |  | 8         | 16.00        | 16        | 32.00        |
| Jan 3 Wed                | Y         |  |  | 8         | 16.00        | 16        | 32.00        |
| Jan 4 Thu                | Y         |  |  | 8         | 16.00        | 16        | 32.00        |
| Jan 5 Fri                | Y         |  |  | 8         | 16.00        | 16        | 32.00        |
| Jan 6 Sat                | N         |  |  |           |              |           |              |
| Jan 7 Sun                | N         |  |  |           |              |           |              |
|                          |           |  |  | 80.00     |              |           | 160.00       |

### **Default Shift Profile**

You can apply a default shift profile to every working day within the year. In addition, you can allow for multiple resources or additional time by specifying a standard and maximum capacity factor. These factors uplift the number of hours defined by the profile per day by the required amount.

The number of hours per day are shown as effective hours. If the shift profile is made up of variable shift lengths, then effective hours may vary.



*Tip: The advantage offered by this method is that, if there are changes in shift length, you simply update the shift profile rather than updating each machine.*

| Production Calendar - 01 |           |                       |                 | Machine A          |              |                    |              |
|--------------------------|-----------|-----------------------|-----------------|--------------------|--------------|--------------------|--------------|
| Date                     | Work Day? | Default Shift Profile | Effective Hours | Default Std Factor | Std Capacity | Default Max Factor | Max Capacity |

|           |   | 1 |   | 1.23 |       | 2.45 |       |
|-----------|---|---|---|------|-------|------|-------|
| Jan 1 Mon | Y |   | 8 |      | 9.84  |      | 19.60 |
| Jan 2 Tue | Y |   | 8 |      | 9.84  |      | 19.60 |
| Jan 3 Wed | Y |   | 8 |      | 9.84  |      | 19.60 |
| Jan 4 Thu | Y |   | 8 |      | 9.84  |      | 19.60 |
| Jan 5 Fri | Y |   | 8 |      | 9.84  |      | 19.60 |
| Jan 6 Sat | N |   |   |      |       |      |       |
| Jan 7 Sun | N |   |   |      |       |      |       |
|           |   |   |   |      | 49.20 |      | 98.00 |

### **Shift/Capacity Profile**

You can allocate one standard shift profile as the default, together with specific shift profiles for those days that deviate from standard.

The example in the table below allocates three shift profiles to Machine A:

| Shift Profile | Description of Shift                         |
|---------------|----------------------------------------------|
| 1             | Standard 8 hour shift                        |
| 2             | Friday shifts with an earlier finishing time |
| 3             | Shift pattern for weekend working            |

In addition, you can apply a standard and maximum capacity factor to each day of the week. For example, you may wish to cater for fewer machines running at weekends. Similarly, you can set up a default code for days when nothing specific is required. Also, you can use effectivity dates to control the number of hours per day.

This method provides the most exacting control on capacity calculations. Again, by changing the profiles you change the daily effectivity hours. Capacity factors, however, should be changed at machine level.

| Production Calendar - 01 |           |               |                 | Machine A  |              |            |              |
|--------------------------|-----------|---------------|-----------------|------------|--------------|------------|--------------|
| Date                     | Work Day? | Shift Profile | Effective Hours | Std Factor | Std Capacity | Max Factor | Max Capacity |
| Jan 1 Mon                | Y         | 1             | 8               |            | 8.00         | 2.34       | 18.72        |
| Jan 2 Tue                | Y         | 1             | 8               |            | 8.00         | 2.56       | 20.48        |
| Jan 3 Wed                | Y         | 1             | 8               | 1.23       | 9.84         | 2.56       | 20.48        |
| Jan 4 Thu                | Y         | 1             | 8               |            | 8.00         | 2.56       | 20.48        |
| Jan 5 Fri                | Y         | 2             | 6               | 1.1        | 6.60         | 1.98       | 11.88        |
| Jan 6 Sat                | N         | 3             | 4               |            | 4.00         |            | 4.00         |
| Jan 7 Sun                | N         | 3             | 4               |            | 4.00         |            | 4.00         |
|                          |           |               |                 |            | 48.44        |            | 100.04       |

## Bundle Tickets

A bundle ticket attaches to each bundle (bag, garment etc.) on a production order.

Style can print bundle tickets automatically, for example at order release. Otherwise, you can request Style to print bundle tickets via a menu option.

An example of a bundle ticket is given below:

| Bundle Ticket              |                                 |
|----------------------------|---------------------------------|
| Style NJACKET Colour SKY   | Blue Size 36 *REPRINT*          |
| New Jersey Jacket          | Order W000212                   |
| Ticket Jacket fronts       | Quantity 12                     |
| Bundle No. 0001516         | Due Date 12/05/96               |
| Lot Number JBAD44          | Next Machine MK12 Next Dep MAKU |
| 0020 First front operation | 0030 Second front operation     |
| Optional bar code          | Optional bar code               |
| 000015160020 6.2750        | 00015160030 3.1575              |

The bundle ticket includes basic data such as:

- Production order number.
- Style.
- Bundle number.
- Colour and size of garments in the bundle.

An operation stub is attached for each operation that is to be performed on the bundle. This stub includes the following information:

- Operation sequence.
- Description.
- Barcode.
- Standard minutes.

The bundle may also include a bar code for speed and accuracy of feedback.

## Bundle Ticket Types

Within Style Production, you can create bundle tickets for different purposes. The following circumstances require different types of bundle tickets:

- Bundles with varying quantities.
- Dividing the route into sections.
- Parallel processing.

Examples are given below.

### **Bundles with varying quantities**

When producing socks, different bundle ticket types may be issued for:

- Knitting (120 pieces per bundle).
- Dyeing (750 pieces per bundle).
- Finishing (240 pieces per bundle).

### **Dividing the route into sections**

In a cut and sew environment, a cutting instruction may be issued to a style. Depending on the mix of sizes and cloth widths, the actual quantities cut may vary from the instruction.

Therefore, all following operations need to be based upon the actual cut quantities and not the original order quantities.

### **Parallel processing**

When producing a shirt, separate components may be produced simultaneously on different machines in various factories.

| Parallel Operations |          |               |          |
|---------------------|----------|---------------|----------|
| Operation           |          | Bundle Ticket | Duration |
| 0010                | CUT      | n/a           | 2 days   |
| 0020                | Front 1  | FRT           | 1 day    |
| 0030                | Front 2  | FRT           | 2 days   |
| 0040                | Front 3  | FRT           | 1 day    |
| 0050                | Back 1   | BCK           | 1 day    |
| 0060                | Back 2   | BCK           | 2 days   |
| 0070                | Back 3   | BCK           | 1 day    |
| 0080                | Sleeve 1 | SLV           | 1 day    |

|      |          |     |        |
|------|----------|-----|--------|
| 0090 | Sleeve 2 | SLV | 1 day  |
| 0100 | Cuff     | CUF | 1 day  |
| 0110 | Make up  | MUP | 7 days |
| 0120 | Pack     |     | 1 day  |

Fronts, backs, sleeves and cuffs may be produced separately and gathered together at an assembly point. In this case:

- A separate bundle ticket can be attached to each bundle of fronts (FRS), backs (BCK), sleeves (SLV) and cuffs (CUF).
- The operations associated with each group of bundles (that is, fronts, backs etc.) can be scheduled to run in parallel.
- These operations can then be brought together at a marry up (MUP) point.



*Note: To use parallel processing, you must use bundle tickets that have had the Parallel Ticket Type set to 1 on the Maintain Bundle Ticket screen.*

| Parallel Processing |                 |                |                 |                |              |
|---------------------|-----------------|----------------|-----------------|----------------|--------------|
| Day                 | Operations      |                |                 |                |              |
| 1                   |                 |                | 0010<br>Cut     |                |              |
| 2                   |                 |                | 0010<br>Cut     |                |              |
| 3                   | 0020<br>Front 1 | 0050<br>Back 1 |                 |                |              |
| 4                   | 0030<br>Front 2 | 0060<br>Back 2 |                 |                |              |
| 5                   | 0030<br>Front 2 | 0060<br>Back 3 |                 | 0080<br>Sleeve |              |
| 6                   | 0040<br>Front 3 | 0070<br>Back 3 |                 | 0090<br>Sleeve | 0100<br>Cuff |
| 7-13                |                 |                | 0110<br>Make up |                |              |
| 14                  |                 |                | 0120<br>Pack    |                |              |

By using different bundle ticket types for different operations, further analysis groups operations by ticket type to determine the actual number of processes run in parallel. The critical path through these operations is the longest lead time associated with a grouped parallel ticket type.

# Style Production Routes

Style Production provides an extremely versatile method of defining style production routes.

The design of the application has been influenced by the need to link materials and operations in order to obtain an integrated model of product and process. This integrated structure allows you to plan, produce and cost products with great precision and control.

For each style production route, you define:

- **Route.** A series of consecutive operations in which materials are transformed into a finished product, to be received on completion into a stockroom or warehouse.
- **Bill of materials.** A list of materials. Link each material to a specific operation, in order that MRP can plan when you need the material.

Each style must have at least one production route.

However, you can set up multiple routes for one style. Examples of when you may require more than one route for a style are given below:

- To contract out certain operations.
- To maintain separate costing and planning routes.
- To define summary route/BOMs for rough cut capacity planning.
- To take account of capacity constraints or alternative processes.
- To accommodate different batch sizes where economies of scale may influence the routing procedure.
- To model different costing strategies.



Tip: Routes are the fundamental building blocks of Style Production. Invest as much time and effort as possible to ensure that they are accurate, complete and kept up-to-date.

## Building A Style Production Route

You can build a route/BOM by:

- Inputting operations manually.
- Copying in operations from groups of standard operation that you set up beforehand.
- Copying in operations from existing routes for the same style or a different style.

Once copied, you can maintain operation details separately for the copied-to product.

Within Style Production, you define production routes at style level. By doing so, you save a lot of time not having to define a production route for every style variant.

You specify which, if not all, of the style's colours and sizes you wish to produce on a particular route. If an individual colour/size or group of colours/sizes uses a variant of the standard route, you can either build a separate route or you can specify variant operations on the same route.

You can define lot size routes that allow you to define product specifications in terms that different functions within your organisation will find appropriate. For example, different roles within the company may build different routes as shown below:

- Accountants in terms of the style's economic production batch size or economic order quantity (EOQ).
- Production planners in terms of the optimum production batch size.
- Material planners in terms of the natural replenishment quantity for the style.

You can control these different views of the style by specifying designated routes for material planning and style costing to ensure consistency of calculation.

Because product specifications and process methods are logically tied together, information is shared between functional areas. For example, industrial engineers and product designers can be encouraged to work together when developing a new style route, and a production design is more likely to result.



Tip: To avoid a situation where numerous unofficial routes/BOMs are held independently of one another, ensure that every functional department has a stake in the route data. This should encourage communication and a greater cohesion between departments, thereby promoting the benefits of information sharing.

### **Standard Operation Groups**

To save time rekeying similar operation definitions against multiple routes, you can define standard operation groups.

Standard operation groups may include both primary and variant standard operations. You can copy standard operations as primary or variant operations to a live production route or model planning route.

Through a unique Specification Id, you can retain a link between route operations and the standard operations from which they originate. Then, when you make changes to a standard operation, you can perform a mass update on the route operations.

There is a parameter you can set up to control which fields get updated during a mass update.

### **Extended Text Descriptions**

You can use an extended text description to enter information for each operation. This information may be essential for complex or commercially sensitive processes, where additional technical or procedural instructions are necessary.

- In support of a quality improvement programme, you can arrange for this information to print out on production order documentation.

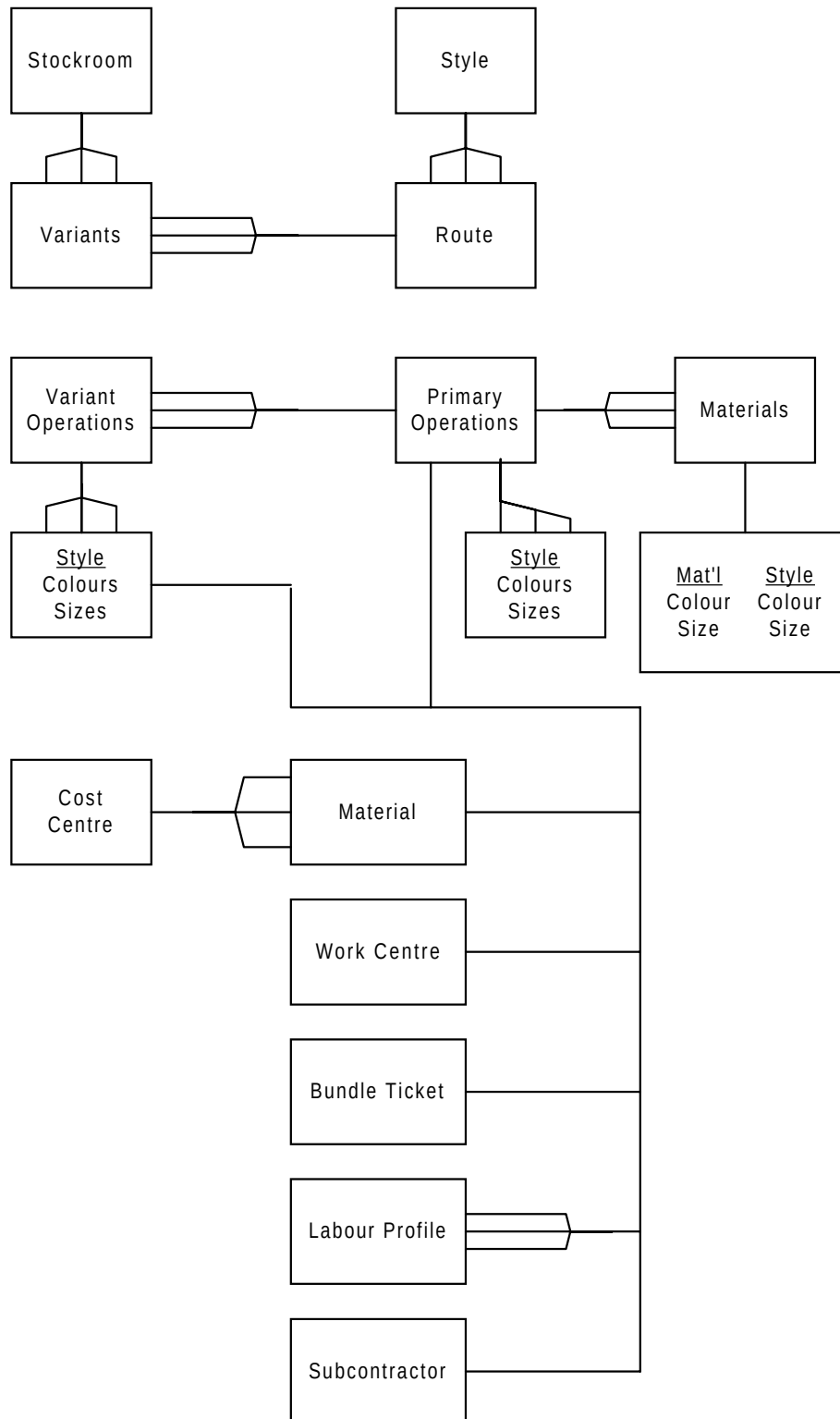


Figure -: Route/BOMs Database Relationships

The above figure shows how a route/BOM relates to materials and resources. A trident indicates a one to many relationship.

## Materials

When you define a route, you provide details of the materials required to make styles and individual style variants.

Styles may require common materials for all style and colour variants. For example, you may make shirts in blue, green and red variants. Each variant may require the same fabric, 100% cotton, but in different colours. In this case, you can use a matrix to relate the material characteristics to the style characteristics. For example, you can specify green fabric for a green shirt.

Alternatively, you can specify different materials for individual colour and/or size variants.

In addition, the material quantity requirements may vary between variants. This is usually associated with larger sizes requiring more material than smaller sizes.

To this end, Style can calculate material requirements automatically for all variants, when you define the quantity requirement for a primary variant and then specify a ratio to determine the remaining variant requirements.

Alternatively, you may wish to specify each requirement individually; with the system defaulting to the standard material requirement entered when first defining the material code against the operation.

### **Updating Material Details On Routes**

To identify and update materials on a route where-used basis, you can use the Mass Replace Materials option. This option saves you the trouble of identifying and updating every occurrence individually.

# Operations

There are two types of operation that you can define on a route:

- **Primary.** A primary operation is an operation used by default to produce the style variants. Style uses primary operations to cost the style.
- **Variant.** A variant operation is an operation that is specific to a certain colour/size variant of a style. The operation details may vary significantly from those defined for the primary operation.

For each primary or variant operation, you need to give the following information:

- Where the operation is carried out.
- How long the operation takes to complete.
- What materials the operation uses.
- How to report completed work.
- The machine at which the operation takes place.

This information is necessary for determining the time-phased load created by the production plan and calculating anticipated production costs.

## **Lead Time And Cost**

For each operation, you specify the standard machine and labour time that it takes to complete the operation. You can base machine and labour time on one of the following units:

- Time taken to produce a standard batch.
- Time taken to produce a quantity of 1.
- Time taken to produce a quantity of 100.
- Time taken to produce a quantity of 1000.

This information is important for two reasons:

- To determine the lead time of the operation for scheduling.
- To cost the operation.

Other factors involved in calculating lead time are:

- Number of machines likely to be available to perform the work. For example, if time is 10 hours but there are 2 machines, then the lead time is only 5 hours.
- Move days factor. This factor indicates any excessive time lost in transferring work from a previous operation. You can add in this factor at machine level by assigning a queue time to the machine.

The cost of the operation is also further qualified by team size or labour profile which determine the number of people required for the operation. Costing routines multiply the number of people by their charge rates to arrive at the labour cost involved in production.

Both costing and lead time values may further be inflated if there is any wastage percentage associated with the operation. This is increased time spent at the operation due to inefficiency. (In capacity planning this may be off-set by machine and labour efficiency factors).

## Variant Operations

The information that you enter for a primary operation applies to all variants of the style. However, this is unrealistic in certain industries that may define a route to cater for a large number of colours/sizes, for example, children through to adults. The time taken to complete one or a batch of styles of children's' sizes is less than that for adults.

To record varying operation details for specific colours and/or sizes, you can set up one or more variant operations. Variant operations run parallel to a style route operation.

From a primary operation, you can define one or more variant operations for certain colours or sizes or colour/size combinations.

Use variant operations to specify:

- Alternative standard production times.
- A different machine.
- A different labour profile (that is, skill mix).
- Changes to required materials, covering quantity per size and matching the colour of the style to that of the material.

## Calculating Operation Lead Time

Operation lead time is the time taken to complete an operation. Lead time is critical for the following functions:

- Scheduling and planning. Style uses the lead time for each operation to determine when to start a production order in order to achieve the due date. This process involves fitting the number of hours required into the number of hours available at the appropriate machine on each working day.
- Capacity loading. Style reviews the hours available per day with that required to progress an order.
- Costing. Style multiplies the machine lead time and the labour lead time by a rate to obtain a cost.

To calculate lead time for an operation, Style uses the following formula:

$$\text{Operation Lead Time} = \text{Queue Time} + \text{Operation Duration} + \text{Move Time}$$

where:

**Queue time** Is the time that completed WIP inventory from a previous operation must wait before work can start on the current operation.

**Operation duration** Is the time that WIP is worked with or on during the current operation.

**Move time** Is used generally to indicate long inter-operational delays, for example moving work between sites.

| Standard Operation Lead Time |            |                    |           |                  |         |                 |
|------------------------------|------------|--------------------|-----------|------------------|---------|-----------------|
| Op                           | Queue Time | Operation Duration | Move Time |                  |         | Total Lead Time |
|                              | Hrs        | Hrs                | Days      | Std Shift Length | Hrs     | Hrs             |
| 10                           | 2.0000     | 4.0000             | 1.000     | 8.000            | 8.0000  | 14.0000         |
| 20                           | 3.0000     | 5.0000             | 0.000     | 8.000            | 0.0000  | 8.0000          |
| 30                           | 4.0000     | 6.0000             | 3.000     | 8.000            | 24.0000 | 34.0000         |
| 40                           | 2.0000     | 3.0000             | 0.000     | 8.000            | 0.0000  | 5.0000          |
|                              |            |                    |           |                  |         |                 |
|                              | 11.0000    | 18.0000            | 4.0000    |                  | 32.0000 | 61.0000         |

## Operation Duration Time

Operation definitions are very flexible in order to mirror the wide variety of methods that may be employed to calculate duration time.

For the purposes of scheduling, operation duration may not be straightforward. The work load may be split between more than one machine.

In addition, it is possible to enable a series of operations to run concurrently (that is, parallel processing) that will reduce the actual lead time dramatically as opposed to completing each operation in turn. However, this does not affect the duration time of the operation itself; nor does it affect costing routines as machine and operator time requirements remain the same.

Standard operation duration depends firstly on manually keying in the following information per operation:

- Set up labour time.
- Run labour time.
- Run machine time.

The exact meanings of these times, entered in hours or minutes depending on the company default requirement, are not necessarily immediately obvious. For example, if you enter a run labour time of 1 hour at an operation, what does it mean? What does your labour produce in that time?

In order to clarify this, a series of time basis codes are available. These codes allow you to clearly indicate what exactly that 1 hour means. You must enter one time basis code per operation, but each operation is independent in its choice of codes.

| Time Basis |                 |                          |           |               |            |             |
|------------|-----------------|--------------------------|-----------|---------------|------------|-------------|
| Lot        | Time Basis Code |                          |           | Set Up Labour | Run Labour | Run Machine |
|            | Description     | Batch Qty                | No.       |               |            |             |
|            |                 |                          |           | Hrs           | Hrs        | Hrs         |
| 1000       | 0               | Time per parent lot      |           | 1.1000        | 12.0000    | 12.0000     |
| 1000       | 1               | Time each                |           | 1.1000        | 0.0050     | 0.0065      |
| 1000       | 2               | Time per 100             |           | 1.2300        | 2.3330     | 1.9800      |
| 1000       | 3               | Time per 1000            |           | 1.3400        | 4.5600     | 3.5000      |
| 1000       | 4               | Quantity per hour (n/a)  |           |               |            |             |
| 1000       | 5               | Fixed time operation     |           | 1.4500        | 7.8000     | 12.0000     |
| 1000       | 6               | Time per operation batch | 500.000 2 | 1.3300        | 1.6780     | 0.9600      |

Another point to consider when calculating operation duration is: do set up labour time, run labour time and run machine time run alternately or are they concurrent?

You may for example, have a run labour time of 5 hours and a run machine time of 7 hours; but the labour time is a 5 hour requirement within the 7 hours of machine time.

Your operation duration time would therefore be set up time plus 7 hours. However, you would still need the run labour time to cost the operation correctly.

Or, for example, you may have a set up time of 2 hours, a run labour time of 3 hours and a run machine time of 4 hours; where the labour component must be carried out before the machine component can start. In this circumstance, you would want an operation duration time of set up plus run labour plus run machine.

To determine exactly how to combine the three time components, Style provides a series of duration basis codes:

| Duration Basis   |                 |                     |                      |                     |                                   |                                               |
|------------------|-----------------|---------------------|----------------------|---------------------|-----------------------------------|-----------------------------------------------|
|                  |                 | Duration Basis Code |                      |                     |                                   |                                               |
|                  |                 | 0                   | 1                    | 2                   | 3                                 | 4                                             |
| Run Machine Time | Run Labour Time | Set Up Only         | Set Up + Run Machine | Set Up + Run Labour | Set Up + Run Labour + Run Machine | Set Up + greater of Run Labour or Run Machine |
| Hrs              | Hrs             | Hrs                 | Hrs                  | Hrs                 | Hrs                               | Hrs                                           |
| 2.00             | 2.00            | 1.00                | 3.00                 | 3.00                | 5.00                              | 3.00                                          |
| 10.00            | 7.00            | 2.00                | 12.00                | 9.00                | 19.00                             | 12.00                                         |
| 3.50             | 1.25            | 1.50                | 5.00                 | 2.75                | 6.25                              | 5.00                                          |
| 20.00            | 15.00           | 3.00                | 23.00                | 18.00               | 38.00                             | 23.00                                         |

## Wastage

You can increase the duration time in order to compensate for expected wastage. This increases the WIP inventory requirement at each operation in order to achieve the operation target.

This wastage % is specified at operation level.

| <b>Wastage</b> |              |              |         |                           |                 |                           |
|----------------|--------------|--------------|---------|---------------------------|-----------------|---------------------------|
| Run Qty        | Std Lot Size | Std Duration | Wastage | Cumulative Wastage Factor | Inflate Run Qty | Inflate Standard Duration |
| 100            | 100          | Hrs          | %       |                           |                 | Hrs                       |
| Operation 30   |              | 5.0000       | 20      | 1.25                      | 125.0           | 6.2500                    |
| Operation 20   |              | 8.0000       | 5       | 1.32                      | 131.6           | 10.5263                   |
| Operation 10   |              | 3.0000       | 10      | 1.46                      | 146.2           | 4.3860                    |
|                |              |              |         |                           | w2              |                           |
|                |              | 16.0000      |         |                           |                 | 21.1623                   |

## Subcontract Operations

Certain processes involved in the production of styles may be carried out off-site at subcontractors. You can include these processes as operations on route/BOMs.

Style Production makes it easy to track work sent to and received from subcontractors. You can track and monitor the:

- Movement of materials to a subcontractor.
- Transfer of WIP inventory to a subcontractor.
- Progress of WIP inventory at a subcontractor.
- Receipt of WIP inventory from the subcontractor.

The steps involving subcontracted operations are as follows:

1. Set up the system.
2. Set up Route/BOM.
3. Release the production order.
4. Create purchase orders for all subcontracted operations.
5. Allocate lots.
6. Issue materials.
7. Report preceding operation.
8. Ship WIP to the subcontractor.
9. Progress at subcontractor (if applicable).
10. Receive from subcontractor.
11. Create GRN for purchase order.

### **System Set Up**

Subcontracting depends upon Style Purchase Management and Accounts Payable being live. Each subcontractor must be a valid supplier. Purchase orders may be raised automatically for all subcontracted operations.

You must define a subcontractor stockroom on the Organisational Model for holding all material balances for materials required by the subcontractor to complete the operation(s).

Set up the Shipper Tracking flag under the Organisational Models option to do one or both of the following:

- Track the movement of WIP inventory and/or materials through a shipper number. Style assigns a number to each shipment. You can, however, use the same number to track amounts of WIP from more than one production order or multi shipment within the same order.
- Print shipper notes (that is, despatch notes) to accompany each shipment.

### **Route/BOM Set Up**

You can define primary operations and variant operations as subcontracted operations.

A subcontracted operation cannot be the first operation on a route/BOM. If you need to get round this, create a dummy count point operation at which to register the work-in-progress to ship.

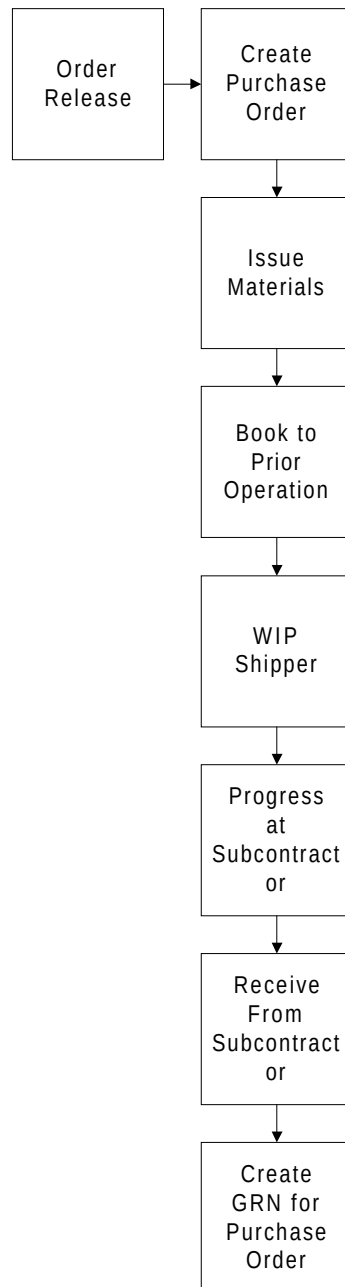
You can record standard labour, machine and set up time to take into account the capacity and loading of subcontracted work when planning.

During material issuing, Style transfers materials to the subcontractor stockroom defined on the Organisational Model. When the operation that the materials are attached to is booked by either the Progress at Subcontractor or Receive from Subcontractor routines, Style backflushes the materials out of the stockroom.

Material balances are held by subcontractor/material within the stockroom.

### **Order Release**

The Release Orders option identifies subcontracted operations on the production route. This option generates purchase orders automatically for all operations where you set the Key Operation field to 2 or 3.



*Figure -: Subcontract Work Cycle*

### **Create Purchase Orders**

Purchase orders are created automatically for the supplier defined on the production route. A purchase requisition is not necessary.

### **Allocate Lots**

If any of the materials to be shipped to the subcontractor are batch or lot controlled, then these materials must be allocated to the production order. Style can then transfer the correct lots to the Subcontractors Stockroom. Later on, these batches/lots can be backflushed.

### **Issue Materials**

Issue any materials required by the subcontractor. Style transfers the materials from the issuing stockroom to the subcontractor stockroom defined on the Organisational Model.

### **Report Preceding Operation**

This is necessary in order to ascertain the quantity of Available WIP to move to the subcontractor.

### **WIP Shipper**

Work-in-progress is now shipped to, and optionally tracked at, the supplier. This option reduces the Available WIP quantity balance at the prior to subcontracted operation, and increases the subcontractor WIP quantity balance at the same operation with the same quantity. Enquiries on subcontractor WIP inventory balances must therefore be made on the prior operation and not the subcontracted operation.

### **Progress At Subcontractor**

If consecutive operations are subcontracted, then it is possible to record bookings at each operation as if the work was being carried out on-site.

Also, subcontractor material issue balances are downdated with material quantities. At the point of receipt, you may override these quantities.

### **Receive From Subcontractor**

On receipt, Style:

- Increases the Available WIP inventory balance at the subcontracted operation by the quantity returned.
- Decreases the subcontractor WIP balance at the prior operation by the same quantity.
- Downdates subcontractor material issue balances with material quantities. At the point of receipt, you may override these material quantities.

Quantities returned from a subcontractor can be received into held status and returned for rework if necessary; subcontractor and prior operation WIP inventory balances are updated appropriately.

### **Create GRN For Purchase Order**

It is customary for Style Purchase Management to carry out this function. Within Style Production, you can report receipts without a GRN (goods received notes) if a different purchase order to that raised by the release procedure is entered during receipts processing.

## How Costing Is Performed

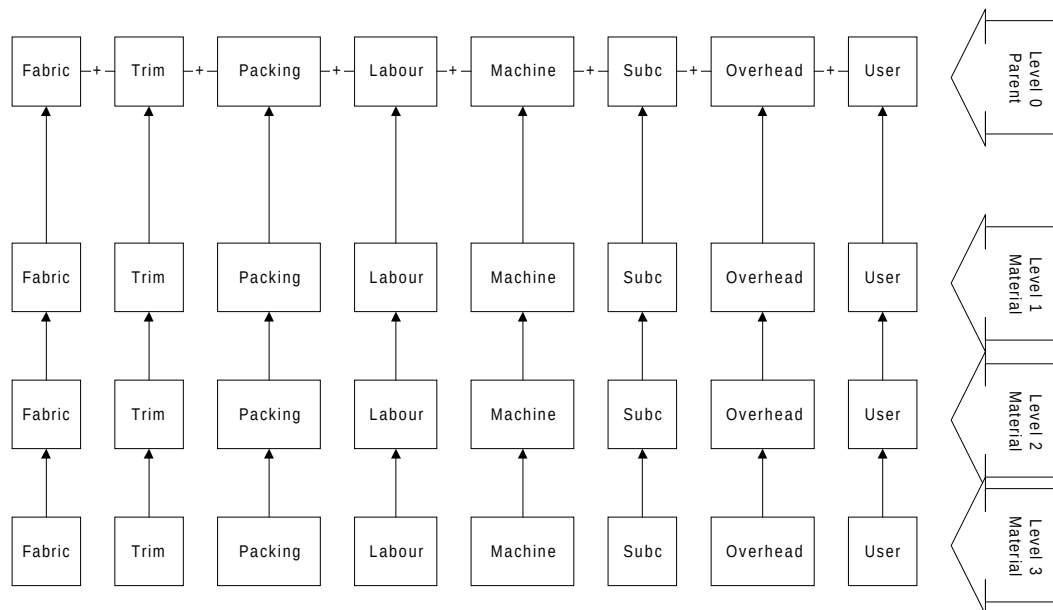


Figure -: Cost Rollup

Cost elements are rolled up into higher level costs to arrive at a full product cost. These cost elements indicate production, materials and user-defined costs at each level in a production route/BOM:

- Production costs comprise labour (set up and run), machine, subcontract and overheads.
- Material costs may be subdivided into fabric, trim and accessory costs.
- User-defined costs may be entered manually; or may comprise exclusively or include landed costs.

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## Chapter 9

### Style Production Planning Overview



# Introduction

Style Production Planning is made up of the following two modules:

- P2      Style Master Production Scheduling
- P3      Style Material Requirements Planning

MRP and MPS are autonomous applications. You may use either application stand alone. However, MRP is normally driven by MPS demand. You can use a separate MRP demand to enhance the requirements for dependent MRP items.

If you set all your styles and materials to be MPS controlled, then you do not need to run MRP. Likewise, if you set all your styles and materials to be MRP controlled, you only need to run MRP. However, MRP does not carry out capacity planning.

If you are a distributor who does not manufacture styles, then you can install either MPS or MRP. However, If you wish to run capacity planning on your purchases, use MPS rather than MRP.



Tip: The advantages of running both MPS and MRP are that you can run MPS, then stop and manipulate the data, before running MRP.

# Planning Models

Master Production Scheduling and Material Requirements Planning utilise planning models.

Planning models define:

- Stockrooms. The planning process considers only materials stocked within, or finished styles received into, stockrooms in the planning model.
- Calendar. This is the production calendar for scheduling work and defining forecasting week numbers.
- Reporting profile. This comprises reporting periods for bucketing planning information on reports and enquiries and MPS/MRP reviews. A reporting profile may be based on a non-linear time horizon. For example, a three month plan might be structured as follows:

|           |                    |
|-----------|--------------------|
| Week 1    | Daily Detail       |
| Week 2-4  | Weekly detail      |
| Week 5-8  | Fortnightly detail |
| Week 8-12 | Monthly detail     |

Use planning models to view the results of alternative plans. You can perform planning runs on different areas of your data and analyse the results of different runs concurrently.



*Note: If you intend to run MRP using the Include Suggested MPS parameter, then you must define identical MPS and MRP models using the same codes, stockrooms and reporting profiles.*

## **The Live Model**

You can define three types of model: centralised, multi-plant and plant. Plants may be defined as live or not live.

To operate MPS or MRP, you must define at least one planning model as the live model in the Company Profile. From the live model, you can change or confirm resulting plans and schedules.

Typically, in a non multi-plant environment, you would have one live planning model. That model, that you would use in regular planning runs to manage your on-going production, would describe the current on-going situation. To simulate other plans, whilst not interfering with the day-to-day operation of your live plan, you could have other planning models.

You can store the output from any MPS run as a revision level. You can change the contents of any revision level and use any revision to generate planning information.

# Master Production Scheduling

The following list shows those processes that are specific to MPS:

- Capacity planning.
- Selection of MPS styles or materials, where you have set the Planning Type to 1 on the Production Details Screen under the Style/Material Details option.
- You can amend suggested production orders, prior to confirmation. For example, you can change quantities, routes, due dates and planning filters.
- You can define a cut-off low level code, below which the run will not look for any occurrence of MPS items.
- You can use revision level codes to save different versions of the master production schedule.
- You can review demand part way through the run, allowing you to effect changes to be incorporated when the run commences.

MPS produces a high level production plan for manufactured items. The system then matches the projected demand for designated products to supply capability, based on the planning models and planning horizons that you define.

Demand is a function of sales orders, forecast, and/or manual entries or a user-defined combination. The ability to supply is a function of the availability of associated materials and resources, as defined within Style Production Definition Management.

Where necessary, the resulting plan suggests amendments to current production orders and/or suggests generating new production and purchase orders. The calculation process for the respective plan generations is simply:

$$\text{Demand less Availability} = \text{Requirement}$$

You can generate a demand forecast by using a simple annual forecast defined within MPS.

MPS is a time based plan of suggestions to meet customer due dates. The plan is derived by the system by working backwards from the customer due date specified, taking account of product lead times, to suggest a latest start date that will achieve completion by the due date.

Production orders may be individually confirmed or confirmed 'en masse' and purchase orders can be automatically passed through to the Requisitioning application for subsequent confirmation and transfer to the Purchase Management application in advance of further MRP planning, if required.

## Capacity Planning

Capacity planning is a unique function within Master Production Scheduling, enabling you to compare available machine and labour capacity against projected work loads.

You can use either the standard planning route/BOMs or specially derived capacity routes.

Through a Capacity Review option, you can change the work load schedule at production order level.

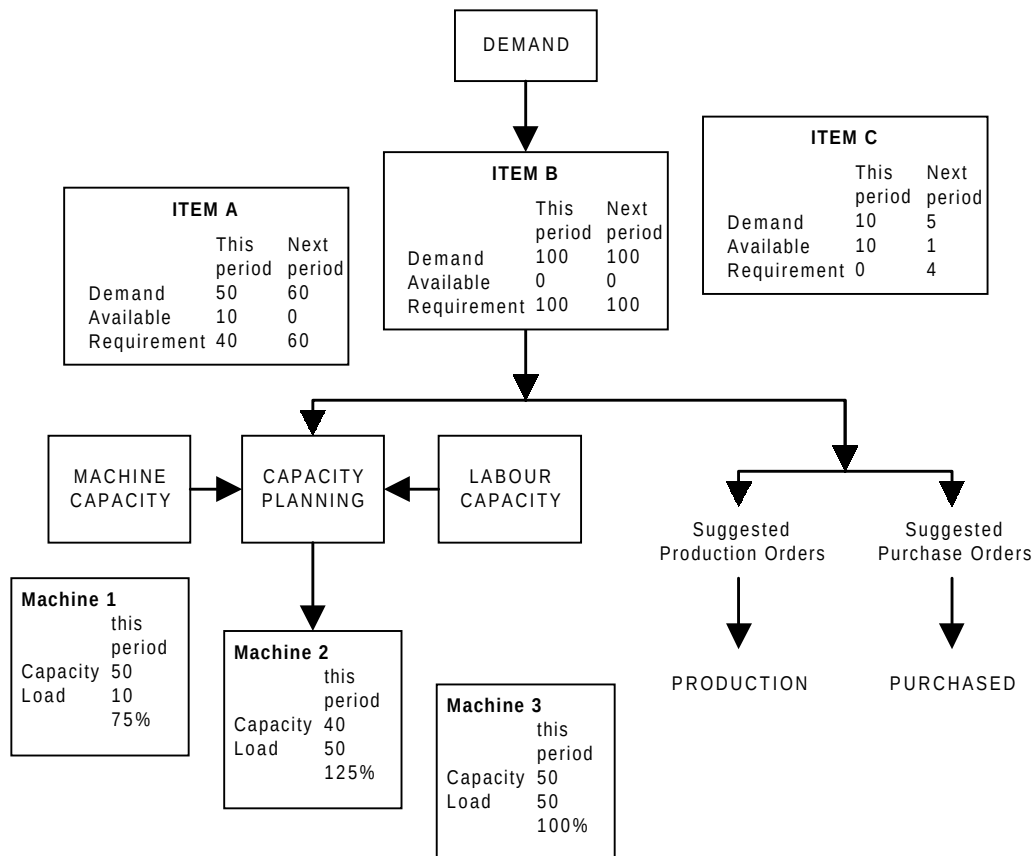


Figure : Master Production Scheduling

MPS incorporates capacity planning to:

- Take account of machine capacities when assessing the demand proposed by MPS plans.
- Determine a loading factor for each machine.
- Determine labour loading.

MPS may use summary routings to determine loading on Style Production resources.

Capacity planning goes through the following process:

1. Uses planning route information to convert demand, in quantities, into the number of hours required at each machine.
2. Uses planned supply dates to schedule those required hours into appropriate weekly production time slots at each machine. Takes the following into account: machine standard capacities; planned down times; non working days.
3. Makes available reports and enquiries from which you can compare the weekly machine/work centre hours required and available within the planning run timescales.
4. From these reports and enquiries, you can search for overload and underload at machines and for labour skills. Then, decide on any action required.

Capacity planning enables finite capacity planning by individual order, but assumes infinite capacity of the facility. It is therefore the responsibility of the planner to complete any fine-tuning of the work load to optimise the loading factor and hence utilisation of production capacity.

# Material Requirements Planning

MRP produces a detailed plan for the purchase of materials.

The following list shows those processes that are specific to MRP:

- Selection of MRP styles or materials, where you have set the Planning Type to 0 on the Production Details Screen under the Style/Material Details option.
- Directly input MPS suggestions into to the material plan.

There is a choice of modes of processing:

- Regenerative. Plan for all materials.
- Selective. Plan for a range of materials rather than all the materials in the planning model.
- Net Change. Consider only those materials that have had any changes to their demand or supply since the last MRP run.

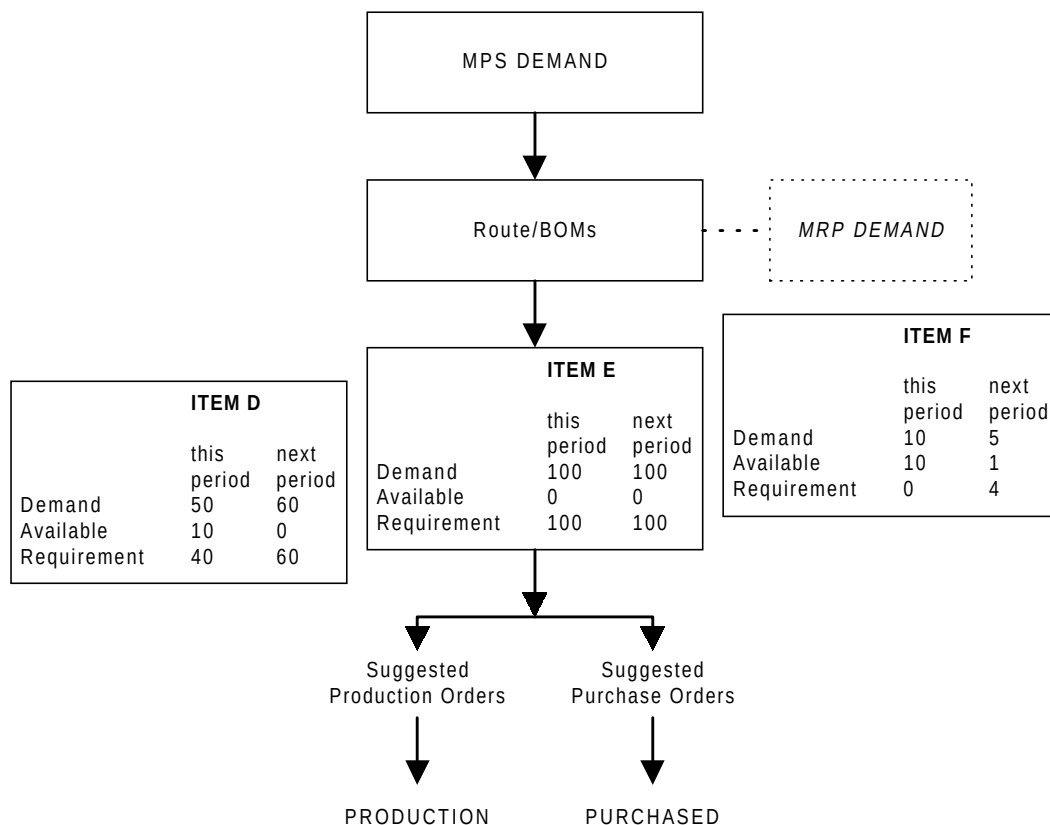


Figure : Material Requirements Planning

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## Chapter 10

### Style Production Control



# Introduction

Use Style Production Control to:

- Record the real world implementation of the production plan.
  - Manage production orders.
  - Make transaction bookings.
  - Monitor the progress of orders through production by logging the status of each order at a machine. The order status may be one of the following:
    - Suggested.
    - Planned.
    - Confirmed.
    - Released.
    - Active.
    - Completed.
    - Cancelled.
  - Record the following events:
    - Issue of materials.
    - Backflushing of bulk issues.
    - Scrap and re-work .
    - Placing unused materials back in stock.
  - Manage machines, work-in-progress inventory (WIP), and inventory stocks.
  - Generate production related reports giving costing and efficiency information.
-  Tip: The accuracy of the reports is dependent on maintaining good discipline in booking operations. Try to balance rigorous discipline with the demands of the production environment.

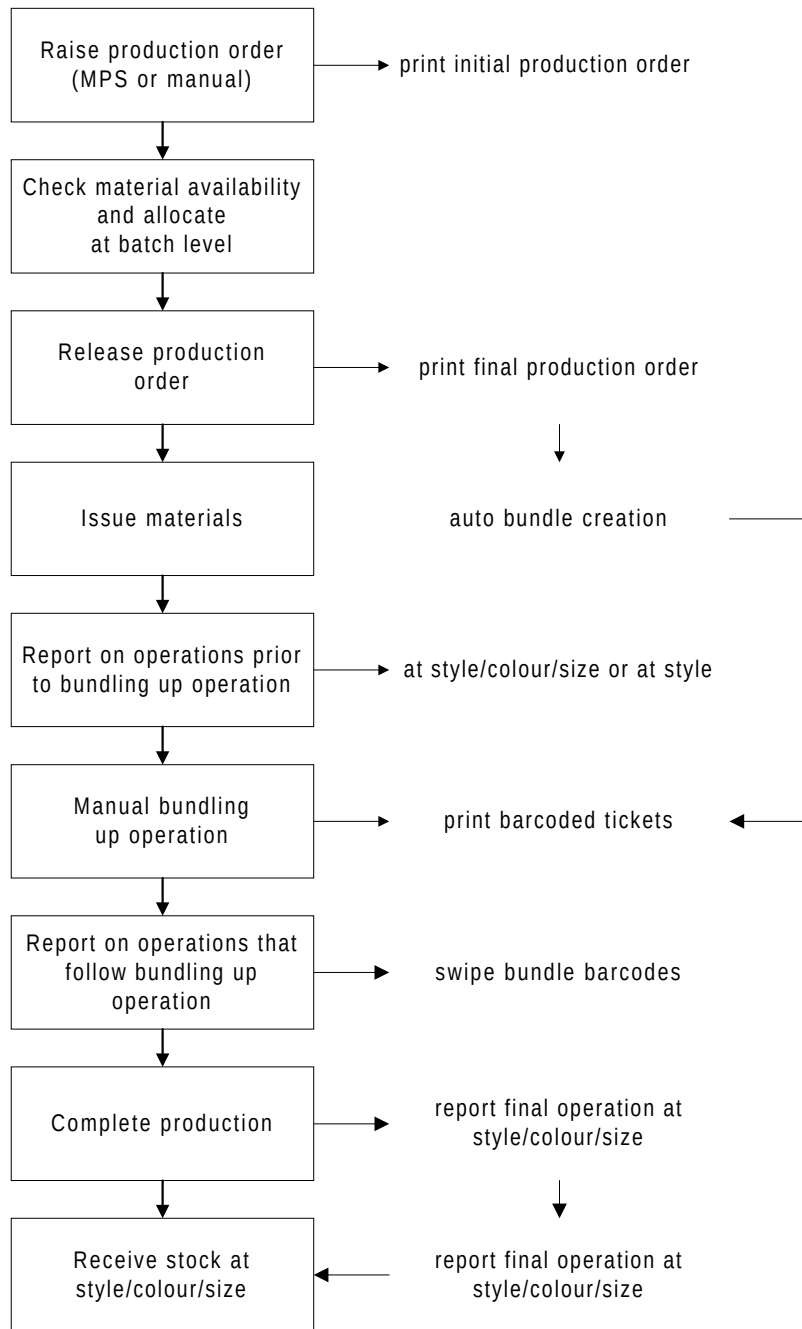


Figure -: Production Control

# Production Orders

A production order is a document that details the manufacture of a specific quantity of one style. The order provides authority for carrying out the manufacturing tasks required.

Production orders originate from two sources:

- MPS/MRP suggested orders. MPS/MRP suggested orders are literally orders with quantities and dates suggested by MPS (Master Production Scheduling application) and MRP (Material Requirements Planning application). Such orders are confirmed within those applications and may be subsequently maintained within Production Control. Suggested orders as such are not visible within Production Control.
- Manually created orders. Manually created orders are orders raised within Production Control. An order may be either of the following types:
  - Standard. Based on a standard pre-defined style route. All MPS/MRP suggested orders are of this type. You can also enter them manually.
  - Amended Standard. Where modifications to a standard route have been made to satisfy a particular demand or circumstance.

The life cycle of an order passes through the stages shown below. Each status is identified on the system either by an alpha code or a numeric code. Relevant codes are shown in square brackets in the subsequent expanded definitions.

- Suggested.
- Planned.
- Confirmed.
- Released.
- Active.
- Complete.
- Cancelled.

## **Suggested [ SUG ]**

This is an order suggested by MPS or MRP processing. A confirmation option changes the status to 'Confirmed' A status of 'Suggested' cannot be entered manually. These orders only appear in the Style Master Production Scheduling and Style Material Requirements Planning applications. The orders are for a specific style and stockroom and have a planning route code, due date and a quantity.

## **Planned [ PLN, 1 ]**

At this point, the order has material and resource requirements based on the selected route/BOM. A planned order represents an intended supply of variants of a finished style. However, at this stage, the order does not create any demand for materials. The orders are for specific styles and stockrooms, and have a planning route code, due date and a quantity. Such orders are shown as 'On

Order' in the Style Inventory Management application. A planned order must subsequently be confirmed.

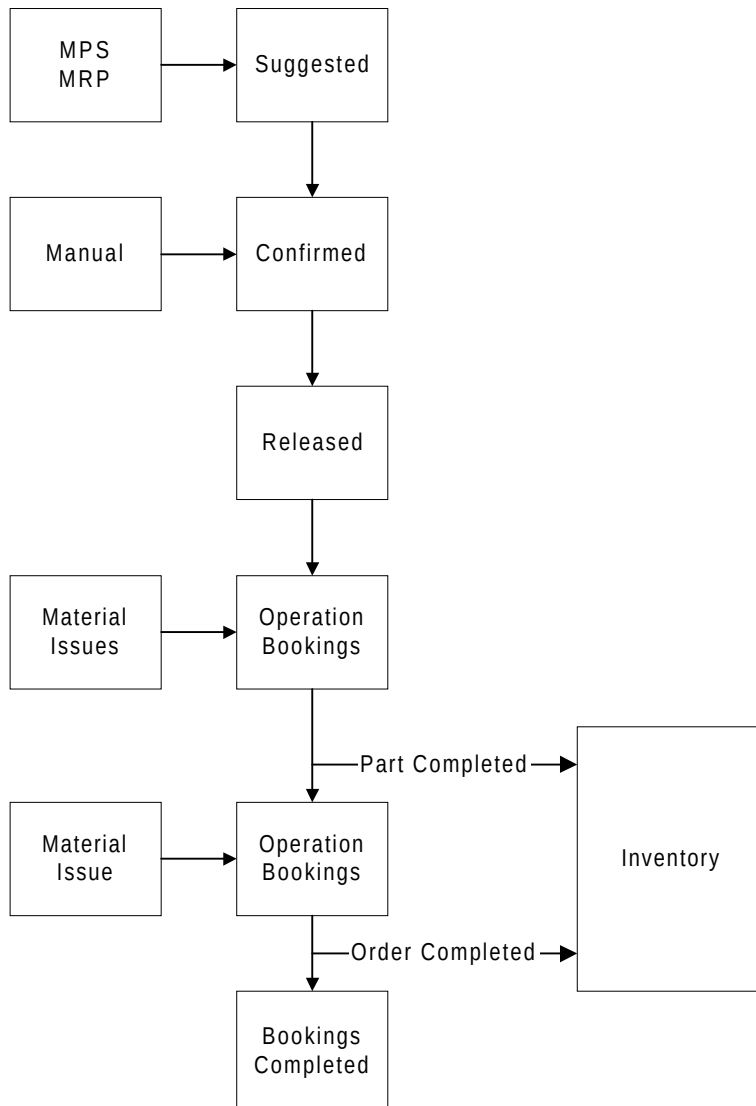


Figure -: Production Order Life Cycle

### **Confirmed [ CON, 2 ]**

Order confirmation causes the allocation of materials in the planned issuing stockroom and/or warehouse if the Allocation of Materials At flag on the Company Profile is set to 1 (confirmation). For standard orders, the system assumes details as defined on the specified route. For amended standard orders, the system uses the standard definition by default, but specific requirements where the standard has been overridden.

The orders have the same characteristics as 'Suggested' orders but with full route/BOM details and resource requirements.

### **Released [ REL, 3 ]**

Order release causes the allocation of materials in the planned issuing stockroom and/or warehouse if the Allocation of Materials At flag on the Company Profile is set to 2 (release) Release prohibits quantity changes

although dates and item detail changes may be made. Issues, order and operator bookings, timesheet and bundle booking and finished item receipts may now be performed.

**Active [ ACT, 4 ]**

The order becomes active as soon as any production activity is recorded against it.

**Complete [ FIN, 8 ]**

An order may be completed after it is released without necessarily reporting any activity. This procedure may be reversed to return the order to 'Active' status, or 'Released' if no bookings were reported. Optionally, an order may be set to completed when the planned quantity is reported, although it is possible to continue to receive more than that planned. Alternatively, it is possible to complete an order without receiving the full planned quantity.

Once an order has reached 'Complete' status it may be archived and purged from the system via the Select Orders to Archive utility. No material issues or bookings are allowed to these orders.

**Cancelled [ CNL, 9 ]**

An order may be cancelled prior to the allocating of materials (that is, at Confirmation or Release) by deleting it. It is then available for purging and archiving. These orders are not accessible and therefore cannot be maintained in any way.

## Maintaining Orders

Production orders originating in Master Production Scheduling and/or Material Requirements Planning may be maintained; together with any orders created specifically within Production Control.

Maintenance facilities are also available for individual orders within:

- Production release option.
- MPS Review (Master Production Scheduling).
- MRP Review (Material Requirements Planning).

Maintenance may require the adjustment of due or start dates, or quantities, or changes to the associated route/BOM. Certain levels of amendment are validated against the order status.

Any maintenance activity which changes the status of the order will generate a transaction.

### **Automatic Calculation of Start Date**

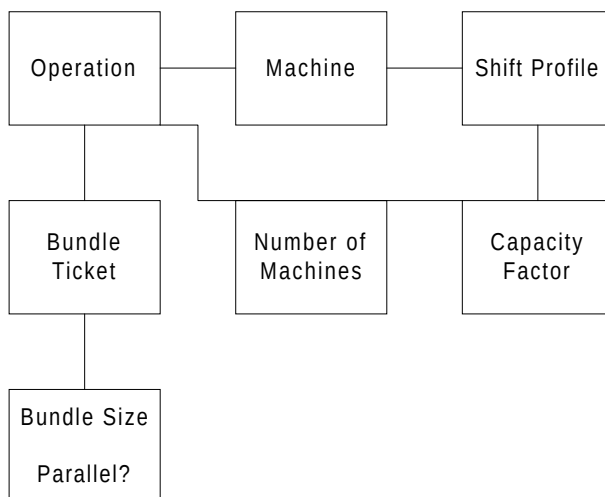


Figure -: Production Order Start Date Calculation

Style Production Control can calculate the start date of an order automatically, as shown below:

#### **Calculate Available Shift Hours Per Day**

Shift profile hours

OR

Total of shift lengths

OR

Default to 8 hours

The available hours are determined by the Machine definition. (See Production Definition Management product guide)

**Calculate Number Of Machines To Be Used In Parallel**

Lowest of:

Number of machines on operations

OR

Order quantity/Default bundle quantity

OR

Capacity factor if shift profiles exist

OR

Shift capacity/Shift length

**Available Hours On Machine For The Order =**

Available shift hours \* Number of machines

## Completing Orders

When all work on a production order at operation and/or order level has been completed, then the order can be set to a status of Completed. At such a status, you can archive the order's details.

Individual operations may be flagged as completed as part of booking options or via the Order Completion facility; such operations may however be re-opened.

Orders again can be flagged as completed as part of the booking options or via the Order Completion facility. Re-opening an operation will re-open the order.

As part of housekeeping options, receipts of WIP which have been recorded against 'unexpected' purchase order numbers and which have no GRN number may be reported. In addition, closed shipper (despatch) note details may be purged from the system.

# Order Documentation

Documentation associated with a production order comprises:

- Production order.
- Bundle tickets.

## **Production order documentation**

The content of the documentation is affected by several Company Profile flag settings:

- The Print Operations on Production Order flag determines whether or not each operation is printed.
- The Print Operation Text on Production Order flag determines whether or not any additional text held against operations is printed.
- The Print Pick Lists flag indicates whether or not materials will be printed.
- The Suppress Backflush Items on Pick List & Production Order flag indicates whether or not bulk issue (backflush) items should be printed on production orders.
- The Suppress Warehouse Items on Pick Lists & Production Order indicates whether or not warehouse controlled items are to be listed on production orders. This flag does not affect pick lists produced by the Warehousing application.

## Archiving Orders

The archive functionality enables you to clear redundant orders so that maximum utilisation of data storage resources can be made.

# Bundles

Bundles enable a more detailed level of tracking work-in-progress. This is necessary because production orders exist for a single style which is broken down into colours and sizes.

Therefore, to simply state that 210 have completed operation 140 is not much use. You need to state which colours and sizes have been completed at the operation.

If bundles have been created then this information is readily available, not just by colour and size, but also bundle number as well.

Bundling up involves gathering a number of pieces of each of the components (for example, fronts, backs and sleeves). Bundling is usually performed at operations such as the cutting table or at a marry up point (ie the point where a series of parallel operations or processes come together).

Bundles can be created as colours only; sizes only; or colours and sizes, allowing bundles to exist for example at the pre-dye stages of production.

The system allows for the printing of bundle tickets which can include bar-coded stubs for each operation. These will be physically attached to the bundle and each stub can then be used to feed back information to progress the work through each operation and also record the operator who performed the work.

You can create bundles either manually or automatically.



*Note: If the Allow Bundle Tracking flag on the Company Profile is set to 0 (no), then you cannot create bundles.*

# Transactions

You can report (that is, book) work carried out against production orders through a number of similar options.

Production order transactions consist of the following options:

- Issue materials.
- Production order booking.
- Operator booking.
- Manual timesheet booking.
- Bundle timesheet booking.
- Subcontractor WIP shipper.
- Progress at subcontractor.
- Receive from subcontractor.
- Transaction enquiry.
- Transaction audit report.

## **Material Issues**

You can:

- Enter planned and unplanned material issues. (A planned issue is one that is derived from the bill of material for the style.)
- Review the current status of orders.
- Identify and issue, if required, substitute materials.
- Issue materials to subcontractors.

## **Production Bookings**

Use production bookings to book times and quantities against one or more operations on a single order.

A production order booking lets you book any event which occurs in the production area. It is designed to be most efficient where both of the following conditions are met:

- All bookings relating to a machine or department are posted consecutively.
- The prime reference to the bookings is the production order number.

Whilst most activities are recorded against a production order, this is not mandatory. Style supports the recording of activities such as cleaning, maintenance and supervision etc. for subsequent activity and performance analysis.

## **Operator Bookings**

An operator booking allows you to book the same type of information as a production order booking. The only difference is that it is most efficient to use

when all bookings for a production order are presented together, and where the activities of individual operators' activities are recorded.

Separate bookings against the same or a range of activity types, operators (or teams) and orders/operations may be actioned whilst retaining the same header details.

### **Manual Timesheet Booking**

A manual timesheet booking allows you to book the same information that can be booked using Production Order Booking.

However, the option is designed to accept data grouped from individual timesheets. You can input individual operators' timesheets where a single elapsed time is entered for a series of activities. For subsequent performance analysis, the actual time is spread pro-rata across all of the activities carried out during the timeframe covered by the timesheet.

Separate bookings against the same or a range of activity types and orders/operations may be actioned. Header details however cannot be maintained within a session.

### **Bundle Timesheet Booking**

Bundle timesheet booking is similar to Manual Timesheet Booking. However, you can use bundle timesheet booking when the operator processes bundles rather than booking against a production order number.

Separate bookings against the same or a range of activity types may be actioned.

### **Subcontract Bookings**

Style provides the following subcontract booking options:

- Subcontractor WIP shipper: Enables you to record the sending of WIP inventory to a subcontractor.
- Progress at subcontractor: Is used to record the progress from operation to operation of WIP inventory at a subcontractor. It is used where multiple consecutive operations on the same production order are subcontracted.
- Receive from subcontractor: Is used to record the receipt of WIP inventory back from a subcontractor.

## Transaction Manager

When you enter a production booking or WIP inventory transaction, the database is not directly updated. Instead, the transaction is placed on the database to await processing by the Transaction Manager. When active, the Transaction Manager regularly reviews the transaction file for any transaction awaiting processing; it then carries out all of the necessary database updating.

The Transaction Manager runs in a subsystem where it exercises a continuing cycle, only becoming dormant after processing all outstanding transactions. It then becomes active once more after 60 seconds and searches for any additional transactions. As the job runs in a subsystem by itself, when activated, it is processed immediately by the system.



Tip: As part of the daily procedures, you should start the Transaction Manager subsystem. Then, initialise the Transaction Manager itself. At the end of the day, stop the Transaction Manager to enable any day-end jobs to be processed.



*Note: If the Transaction Manager is not active, any transactions you enter do not get processed until you re-start the Transaction Manager.*

## Transaction Reviews

Style provides the following transaction review options:

- Transaction enquiry: Provides an on-line capability to review the contents of transactions which have been input to the system.
- Transaction audit report: Provides the ability to produce a report on the transactions which have been input to the system selecting from a range of parameters.

## Activity Codes

All bookings are made against an activity type code that has an associated reporting type that identifies the updates effected by the transaction. In addition to reporting WIP available (good), held and scrap quantities at count point operations, the options are used to book a range of other activities:

- Down time.
- Set up time and materials.
- Indirect overheads.
- Rework.
- Off standard hours.
- Labour and machine time at non count point operations.
- Scrap at non count point operations.

### **Down Time**

To book machine and operator (team) down time, you must enter time against an activity type with a reporting type of 05 (down time). Time and costs associated with down time are reviewed on the Down Time Report

### **Set Up Time And Materials**

To book set up labour time, you must enter time against an activity type with a reporting type of 03 (set up). Any materials consumed specifically by the set up process may be entered by adding an unplanned material issue via the Materials window when booking set up time. If materials are not recorded at this point, they do not contribute to the set up cost. These times and costs are reviewed on the Set Up Cost Report.

### **Indirect Overheads**

Indirect overheads are costs incurred within the production environment that are not directly attributable to production. For example, cleaning. As such, they do not contribute to the cost of items; but if such costs become significant they may need to be covered by the direct overheads defined at cost centre level. These overheads are reviewed on the Indirect Overhead Report.

To book indirect overheads against labour, machines and materials, you must enter time against an activity type with a reporting type of 06 (indirect overhead). Any indirect material costs incurred may be entered by adding an unplanned material issue via the Materials window during indirect overhead time booking.

### **Rework**

Two sorts of rework are catered for:

- Rework held WIP. This is where a quantity has been booked as held, possibly as a result of non conformance to quality standards, but has been reworked to comply, and then made 'good' or available. To record labour and machine time to complete the work, time must be booked against an activity type with a reporting type of 04 (rework held WIP).

- Rework at non count point operation. Although no quantities may be recorded against non count point operations, obviously, time may still be spent reworking inventory that would otherwise be missed by the backflushing process. To record such labour and machine time, time must be booked against an activity type with a reporting type of 11 (rework at non count point operations).

The Scrap and Rework Analysis Report considers both types as grouped rework. However, if you designate different activity type(s) for each reporting type, then they are reviewed independently via the Rework Activity Code Analysis Report

### **Off Standard Hours**

Work booked against production may be referred to as On standard hours. However, in certain environments, especially those that utilise piece work, time given over to training or retraining in an unfamiliar job, is referred to as Off standard hours and needs to be recorded independently of On standard hours.

The Manual Timesheet Booking option identifies such time when reporting operator timesheet performance as any time other than that booked against an activity code with a reporting type of 01 (production reporting) or 05 (down time). Such activities would usually be entered against an activity type with a reporting type of 07 (rework).

### **Labour And Machine Time At Non Count Point Operations**

Although actual material costs at operations may be recorded directly by referencing material issues functions, actual labour and machine time at non count point operations must be entered specifically against the operation.

To record this time, time must be entered against an activity type with a reporting type of 07 (labour, machine at non count point operation) at the non count point operation. System validation checks ensure that no WIP inventory quantities are entered.

### **Scrap At Non Count Point Operations**

To record the scrapping of WIP inventory at non count point operations, you must book the scrap quantity against an activity type with a reporting type of 34 (scrap at non count point operation). This scrap value then contributes to the actual production costs of the order.



*Note: Booking any activity against an order for the first time changes the status of the order to 'Active'*

# WIP Inventory Management

WIP inventory management comprises a series of booking transactions that enable the adjustment of inventory balances.

Inventory balances at each WIP location are maintained as a consequence of recording production bookings. Inventory levels can also change as a result of other events (eg scrapping inventory; carrying out physical inventory checks).

At times, you may need to modify WIP inventory transactions generated by booking options eg due to changes in circumstances or events affecting inventory after it has been booked.

Inventory booked good at an operation may be found to be faulty subsequently, eg inventory may be left under bright lights, causing colours to fade. In this case, use the Hold Inventory option to prevent movement of the inventory to the next operation until the inventory is reworked, released or scrapped. If you find that inventory is so badly affected or damaged that it has to be scrapped, use Scrap Available Inventory.

To release WIP previously held through WIP Inventory Control options or placed directly into held status at booking time, usually as a QA stage, use Release Inventory.

Inventory may be booked to WIP locations other than standard (ie the WIP location expected on the route), due to it being worked on by different machines with differing WIP locations. In this case, it may be necessary to re-locate inventory back to the standard location in order to effect proper costing and generate movement records. To do this, use Transfer Location.

Further options allow you to reallocate WIP from one production order to another and to adjust the inventory balance at a WIP location.



*Note: You cannot manipulate available WIP after the last operation on a route. If you book WIP inventory as held, you have to release the inventory at the last operation.*



*Note: WIP Inventory Management prompts with a matrix to determine the styles/variants comprising the total quantity.*

## **Calculation of Actual Time**

When all productive work against the hours entered has been defined, the system calculates the assumed actual time for each activity. This time is posted on each individual transaction record and is subsequently used in performance analysis. The actual time of is calculated as follows:

$$\frac{\text{Standard Time for Activity}}{\text{Total Standard Time}} \times (\text{Elapsed Time} - (\text{Off Std Hrs} + \text{Down Time}))$$

Where:

Standard Time for Activity - Is the standard time for the activity calculated from the production route.

Total Standard Time - Is the sum of all the standard times for the activities recorded on the timesheet.

Elapsed Time - Is the number of hours entered on the timesheet header.

Off Std Hrs - Is the number of labour hours on the timesheet booked to activities other than those with reporting types of 01 (standard production) or 05 (down time)

Down Time - Any time booked against an activity type with a reporting type of 05 (down time)

# Resource Load Management

Resource load management comprises the following functions:

- Review Machine Load.
- Manage Machine Load.
- Review Labour Load.
- Maintain Machine Status.
- Daily Factory Loading.

Style Production considers two kinds of resource:

- Machines which may be grouped for analysis into work centres.
- Labour which comprises operators grouped optionally into teams, and which may further be analysed to operator/skill level.

Facilities within Resource Load Management enable you to review the current loading on machines and labour and to reschedule where any capacity overload can be offloaded to underloaded resources.

# Reviewing The Performance Of Your Production Facility

In order to review the performance of your production facility, a range of standard reports is available, grouped into three categories:

- Machines.
- Labour.
- Scrap and Rework.

## **Machines**

Although grouped under an umbrella of machine performance reporting, these reports also have labour elements, as the activities covered can often be booked to labour as well as machines:

- Machine Efficiency.
- Machine Utilization.
- Set Up Cost Report.
- Direct Overhead Variance.
- Down Time Report.
- Indirect Overhead Report.

It must also be remembered that the machine definition is open to interpretation, such that a machine need not necessarily represent a mechanical device requiring operators; but may itself be used to represent one or more operators.

## **Labour**

Use the following report to analyse performance with regard to actual and standard hours booked:

- Direct Labour Variance.

## **Scrap And Rework**

These reports enable you to review why WIP inventory has been scrapped and held inventory reworked:

- Scrap and Rework Analysis.
- Scrap Reason Code Analysis.
- Rework Activity Code Analysis.